NASSLI
LED LIGHTS

Giant street lighting device

Application:

Ideal for collector roads, local roads, commercial and industrial complexes, driveway, open areas, public squares, parks and playgrounds, factories, airports.

Feature:

Fixture designed for harsh and hot environments.

Each LED module can be replaced separately. Overload protection, short circuit protection, and no-load protection. Aluminum heat dissipation module has a surface anodic oxidation process to make it resistant and anti-aging fixture.

SPECIFICATIONS

ITEM	DESCRIPTION
Input AC Voltage	90-305 VAC
Frequency	47-63 Hz
Power Factor	>98%
CCT	2700K, 3000K, 4000K, 6500K
CRI	>70%
LED Efficiency	
Power supply	INVENTRONICS
LED Type	Philips
Materials	Aluminum, PC, & Friendly material
Beam Angle	143° x 67°
Life Time	>100,000 Hour
Operating Temp	-20°C to +55°C
Storage Temp	-40°C to +85°C
Certifications	CE, SELV and ISO
Mounting	With Pole
IP Rating	IP66
Dimmable	1...10 V On Request
Emergency	N/A
Warranty	5 Years

Order code	Dimensions(mm)	Power(Watt)	Lumen
NSL-STLM-127W-XXK-SMD-IN-IP66	L420 W320 H27	127	20021
NSL-STLM-153W-XXK-SMD-IN-IP66	L420 W320 H27	153	23972

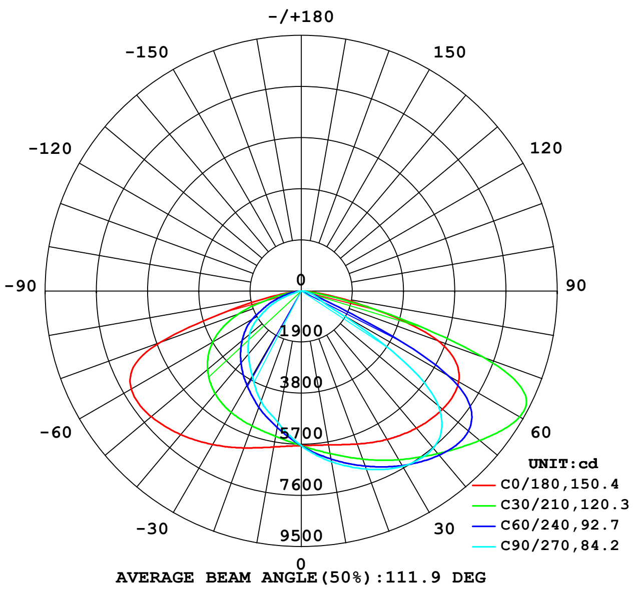
NASSLI LED Factory C.R. 1010328884
www.nassli-led.com.sa Kingdom of Saudi Arabia
Riyadh (Head Office), P.O. Box25320 Riyadh 11466, UN: 920004733

STREETLIGHT PHOTOMETRIC TEST REPORT

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm**		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

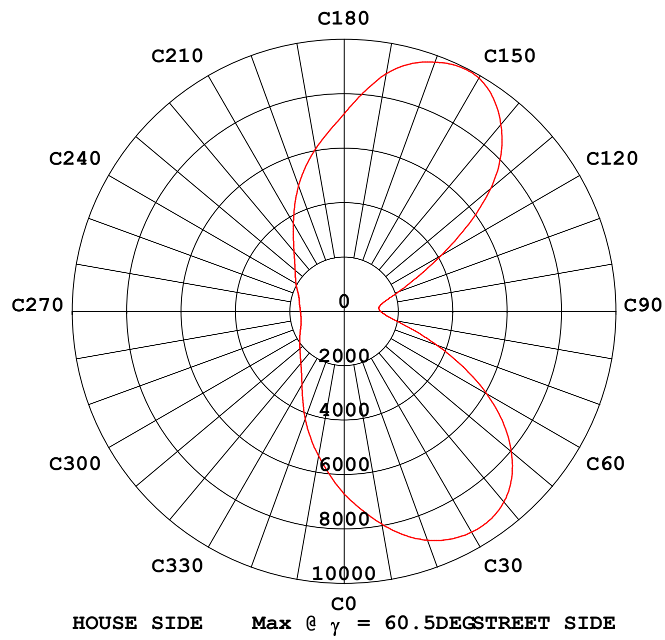
DATA OF LAMP		**PHOTOMETRIC DATA**	**Eff: 154.83 lm/W**		
MODEL	**STREET LIGHT**	**Imax(cd)**	9911	**η street_up(%)**	0.0
NOMINAL POWER(W)	153	**LOR(%)**	100.0	**η street_down(%)**	63.6
RATED VOLTAGE(V)	220	**TOTAL FLUX(lm)**	23972	**η house_up(%)**	0.0
NOMINAL FLUX(lm)	23972	**MAXIMUM @(C,γ)**	150,60.5	**η house_down(%)**	36.4
LAMPS INSIDE	1	**η up(%)**	0.0	**76 FLASHAREA(m2)**	0.05000
TEST VOLTAGE(V)	220	**η down(%)**	100.0	**SLI**	23.853

INTENSITY DISTRIBUTION DIAGRAM
IN C PLANS



AVERAGE BEAM ANGLE(50%):111.9 DEG

MAX INTENSITY CONE SURFACE
DISTRIBUTION DIAGRAM



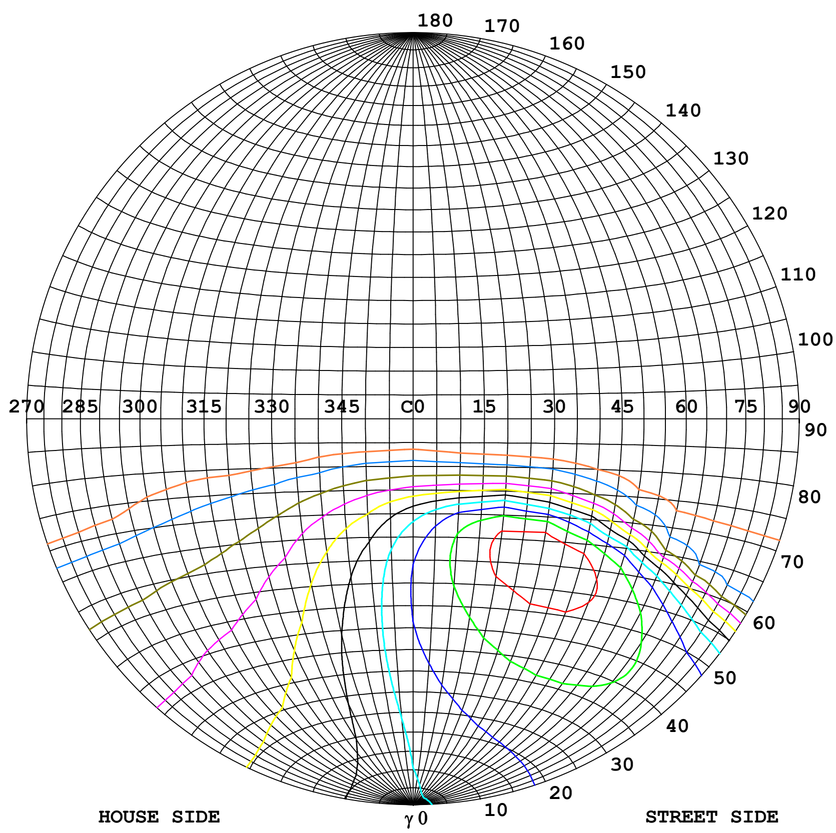
HOUSE SIDE Max @ γ = 60.5DEG STREET SIDE

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

STREETLIGHT ISOCANDELA DIAGRAM

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm		
NAME: NSL-STL-153W-6500-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		



Classification:

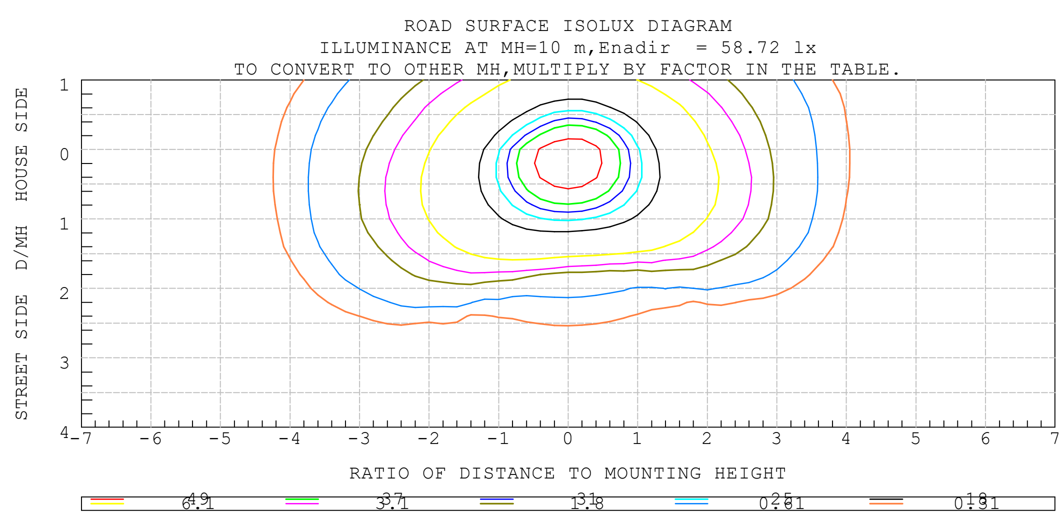
IES:Type II - Short
CIE:Average - Intermediate
IES:Cut-off
CIE:Semi-cut-off
Max.At80:54.65cd/klm
Max.At90:2.349cd/klm
Max.80-90:54.65cd/klm

ISOCANDELA DIAGRAM	
UNIT	cd
Imax=100%	9911
90%	8920
80%	7929
70%	6938
60%	5947
50%	4956
40%	3965
30%	2973
20%	1982
10%	991
5%	496

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

COEFFICIENT OF UTILIZATION CURVE AND ISOLUX DIAGRAM



HEIGHT	FACTOR
5 m	4.000
6 m	2.778
7 m	2.041
8 m	1.563
9 m	1.235
10 m	1.000
11 m	0.826
12 m	0.694
13 m	0.592
14 m	0.510
15 m	0.444
16 m	0.391

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ZONAL FLUX DIAGRAM

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

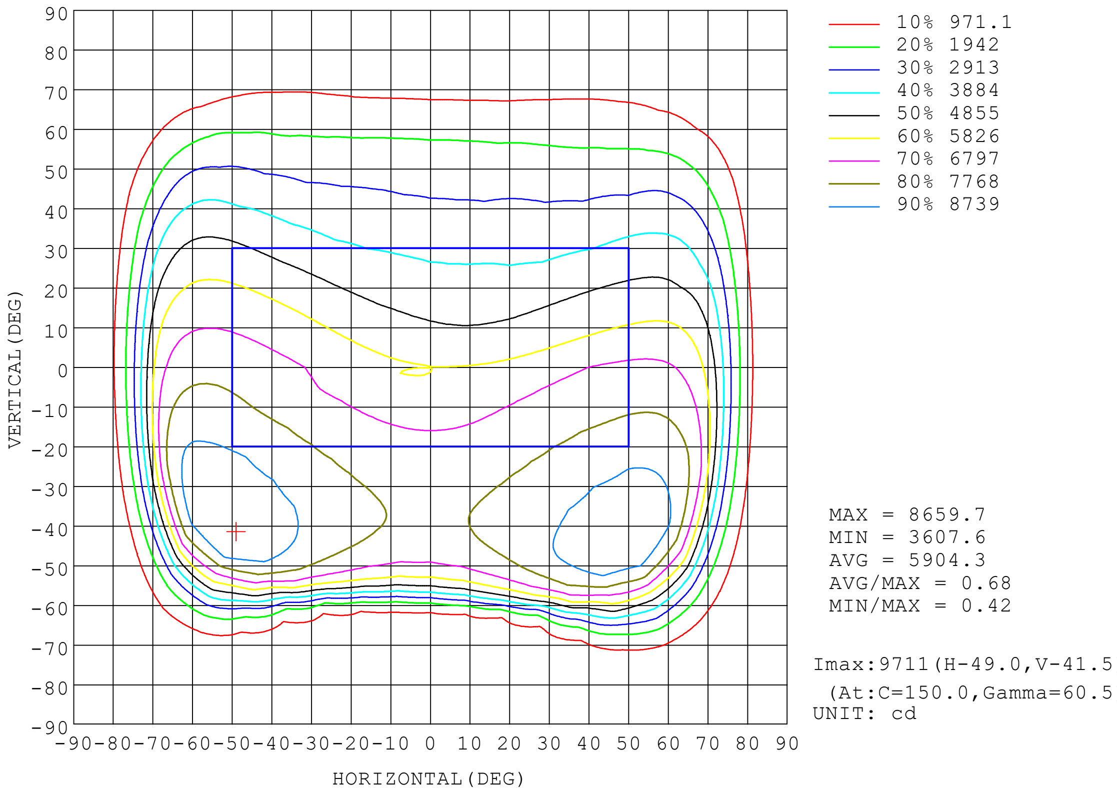
γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	Φ zone	Φ total	%lum,lamp
10	5809	6244	6439	6349	5885	5310	4977	5178	0- 10	549.9	549.9	2.29,2.29
20	6053	6812	7043	6999	6206	4943	4369	4690	10- 20	1653	2203	9.19,9.19
30	6370	7420	7523	7662	6586	4544	3644	4189	20- 30	2753	4955	20.7,20.7
40	6654	8045	7621	8316	6963	4065	3061	3668	30- 40	3793	8748	36.5,36.5
50	6837	8647	6636	8837	7269	3458	2508	3076	40- 50	4662	13411	55.9,55.9
60	6744	8656	1622	8162	7287	2669	1665	2354	50- 60	4976	18386	76.7,76.7
70	5382	3851	560.4	1999	5541	1535	652.7	1331	60- 70	3851	22238	92.8,92.8
80	1310	365.5	200.2	242.8	893.0	264.7	101.5	330.8	70- 80	1562	23800	99.3,99.3
90	53.00	37.43	41.92	46.70	46.01	55.55	42.63	43.37	80- 90	172.1	23972	100,100
100									90-100			
110									100-110			
120									110-120			
130									120-130			
140									130-140			
150									140-150			
160									150-160			
170									160-170			
180									170-180			
DEG	LUMINOUS INTENSITY:cd									UNIT:lm		

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ISOCANDELA DIAGRAM

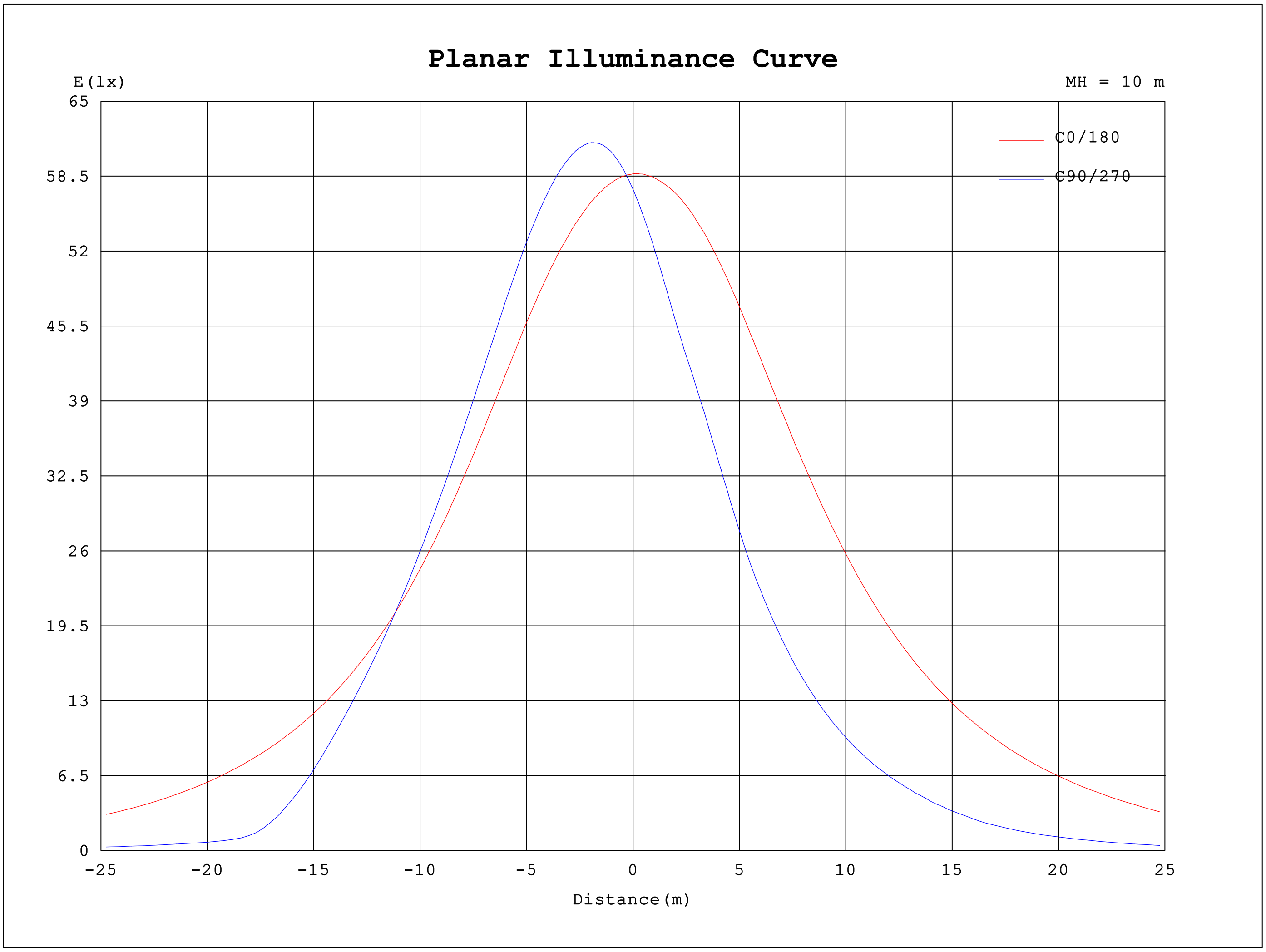
Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm**		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		



C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

Planar Illuminance Curve



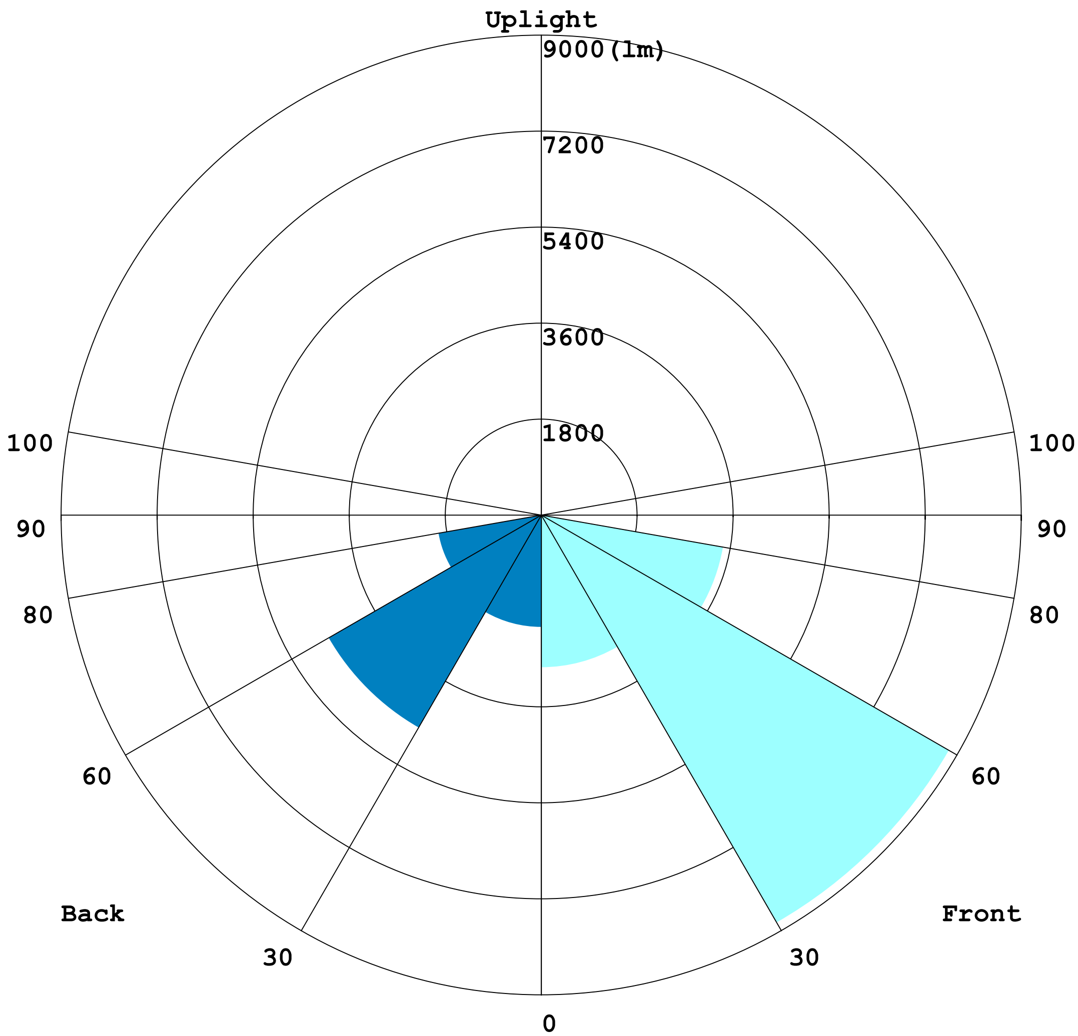
C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LCS REPORT

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

LUMINAIRE CLASSIFICATION SYSTEM(LCS) GRAPH



C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

BUG REPORT

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm		
NAME: NSL-STL-153W-6500-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		

IESNA Luminaire Flux Distribution Table:

Zone	Lumens	Luminaire %
FL - Front-Low(0-30)	2858.2	11.9
FM - Front-Medium(30-60)	8827.5	36.8
FH - Front-High(60-80)	3460.3	14.4
FVH - Front-Very High(80-90)	93.271	0.4
Total Forward Light	15239	63.6

BL - Back-Low(0-30)	2097.2	8.7
BM - Back-Medium(30-60)	4603.4	19.2
BH - Back-High(60-80)	1953.3	8.1
BVH - Back-Very High(80-90)	78.79	0.3
Total Back Light	8732.7	36.4

UL - Uplight-Low(90-100)	0	0.0
UH - Uplight-High(100-180)	0	0.0
Total Up Light	0	0.0

BUG(Back,Up,Glare) Rating	B3-U0-G3

Zone	Downward Lumens	Upward Lumens	Total Lumens
House Side	8732.7	0	8732.7
Street Side	15239	0	15239

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

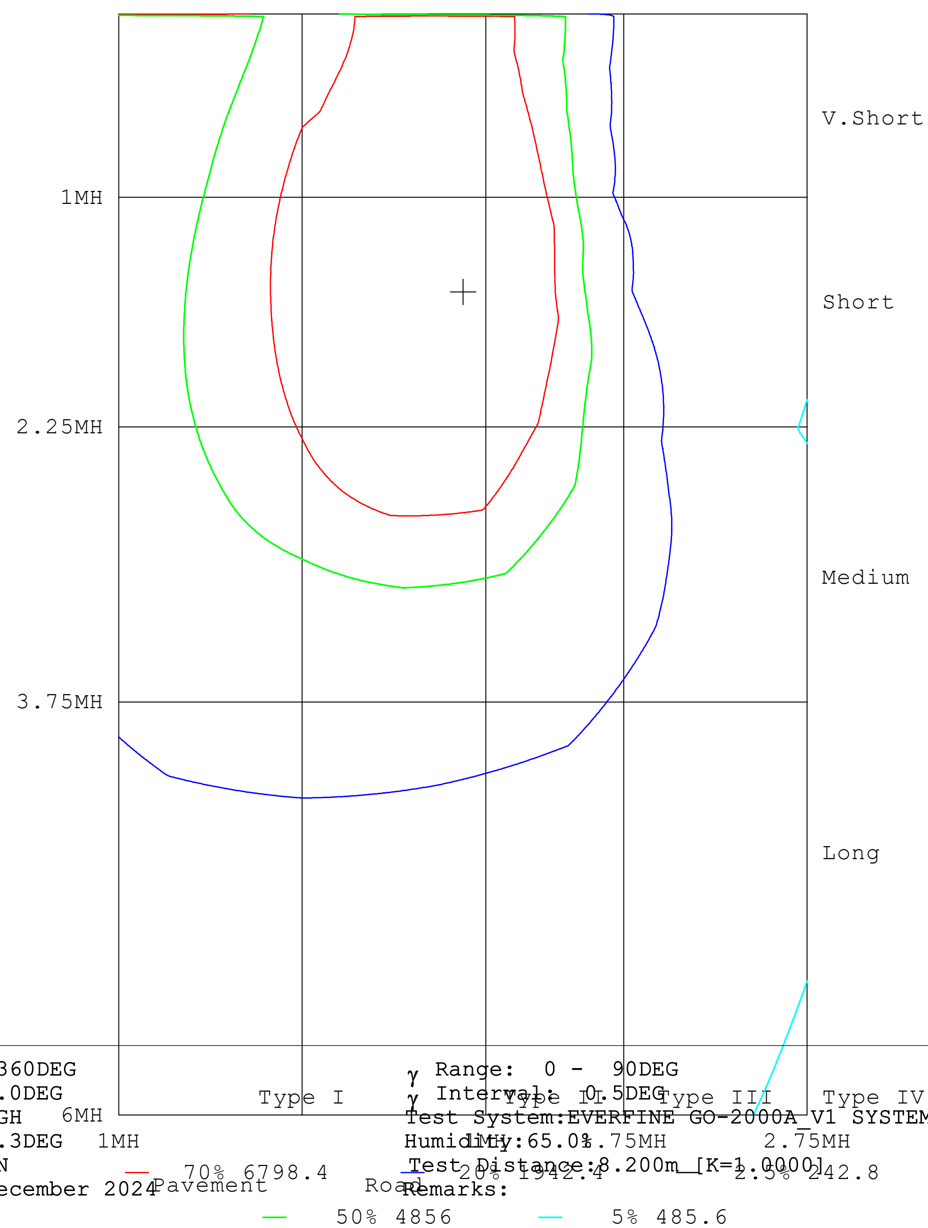
γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ROAD ISOCANDELA REPORT

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz	
UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm	

NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

ROAD SURFACE ISOCANDELA DIAGRAM



C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm**		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

Table--1 **UNIT: cd**

C(DEG) / γ (DEG)	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180
0	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747
5	5754	5815	5872	5925	5970	6012	6047	6077	6097	6110	6111	6104	6086	6062	6026	5984	5931	5872	5788
10	5809	5929	6036	6131	6209	6280	6338	6386	6417	6439	6444	6438	6416	6377	6320	6248	6159	6046	5885
15	5912	6089	6243	6375	6477	6566	6629	6681	6719	6747	6754	6755	6737	6703	6637	6543	6421	6269	6031
20	6053	6288	6488	6650	6766	6858	6923	6975	7011	7043	7052	7060	7052	7029	6968	6866	6716	6517	6206
25	6213	6505	6749	6943	7067	7157	7211	7251	7280	7312	7325	7345	7354	7353	7305	7202	7025	6782	6397
30	6370	6724	7016	7241	7378	7463	7490	7493	7493	7523	7543	7592	7638	7675	7649	7544	7342	7049	6586
35	6517	6943	7291	7552	7702	7768	7744	7691	7633	7637	7674	7777	7887	7986	8004	7906	7681	7328	6781
40	6654	7159	7583	7889	8036	8055	7964	7818	7662	7621	7671	7854	8073	8271	8360	8295	8042	7618	6963
45	6763	7369	7889	8251	8375	8326	8112	7805	7486	7355	7427	7733	8122	8501	8711	8711	8438	7920	7130
50	6837	7580	8235	8644	8732	8561	8095	7484	6905	6636	6716	7221	7904	8615	9059	9152	8882	8242	7269
55	6854	7790	8615	9054	9071	8639	7694	6572	5476	4913	5001	5900	7119	8411	9316	9597	9349	8580	7345
60	6744	7927	8931	9357	9172	8139	6307	4019	2282	1622	1623	2535	4538	7266	9060	9909	9724	8850	7287
65	6353	7827	8918	9174	8207	6104	2421	885	785	773	783	828	1077	3376	7091	9199	9592	8830	6922
70	5382	6903	7706	7173	5365	2331	699	586	563	560	563	585	625	915	3085	6039	7620	7109	5541
75	3253	3714	4128	3436	1905	679	411	386	371	365	365	379	399	446	885	2222	3116	3093	2686
80	1310	1217	1126	896	485	245	213	205	201	200	197	202	201	218	267	507	775	753	893
85	293	198	181	162	110	86.5	84.4	82.2	82.8	79.2	81.2	81.9	82.4	84.4	85.8	106	146	107	136
90	53.0	47.3	42.1	38.9	37.6	37.3	37.7	38.8	40.4	41.9	43.7	45.3	46.4	46.9	46.5	46.1	46.0	46.0	46.0

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.02V I:0.7226A P:154.82W PF:0.9738 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:23972x1 lm**		
NAME: NSL-STL-153W-6500-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

Table--2 **UNIT: cd**

γ (DEG) \ C (DEG)	190	200	210	220	230	240	250	260	270	280	290	300	310	320	330	340	350
0	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747	5747
5	5728	5664	5604	5550	5496	5450	5411	5384	5368	5365	5375	5398	5434	5482	5539	5603	5670
10	5757	5620	5487	5368	5253	5152	5068	5010	4977	4973	4997	5050	5127	5230	5353	5486	5630
15	5836	5621	5403	5207	5025	4877	4761	4687	4661	4665	4690	4741	4847	4996	5184	5400	5635
20	5943	5644	5339	5069	4817	4611	4463	4405	4369	4370	4411	4481	4590	4790	5042	5349	5681
25	6063	5678	5279	4924	4583	4318	4174	4080	4009	4010	4076	4192	4315	4567	4911	5318	5749
30	6179	5710	5208	4754	4334	4042	3889	3744	3644	3647	3730	3896	4048	4330	4770	5289	5825
35	6292	5726	5116	4560	4081	3772	3613	3427	3344	3346	3421	3617	3781	4088	4604	5242	5880
40	6379	5705	4977	4325	3804	3488	3320	3131	3061	3065	3128	3333	3502	3834	4409	5160	5911
45	6434	5641	4787	4055	3503	3189	3007	2849	2788	2797	2853	3039	3210	3555	4179	5032	5900
50	6442	5521	4541	3738	3176	2869	2690	2569	2508	2519	2574	2722	2900	3252	3897	4850	5837
55	6355	5313	4219	3364	2811	2530	2364	2230	2152	2169	2245	2407	2568	2915	3564	4600	5702
60	6112	4966	3795	2929	2408	2105	1856	1727	1665	1688	1759	1958	2177	2531	3164	4248	5448
65	5577	4400	3235	2410	1872	1558	1346	1244	1205	1211	1296	1433	1649	2080	2658	3732	4948
70	4459	3438	2459	1733	1336	1115	905	754	653	713	817	1002	1196	1467	2054	2962	4054
75	2509	2006	1346	1071	844	685	431	212	207	214	253	561	752	957	1231	1901	2703
80	925	744	562	294	235	296	133	102	101	99.6	104	193	337	325	544	808	1202
85	159	105	87.4	81.8	74.3	70.0	58.9	53.2	49.2	46.9	50.4	58.4	68.6	77.3	89.3	168	260
90	51.0	54.5	56.2	56.3	54.8	52.0	48.6	45.3	42.6	40.9	40.4	41.1	42.6	44.2	45.7	46.7	47.1

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

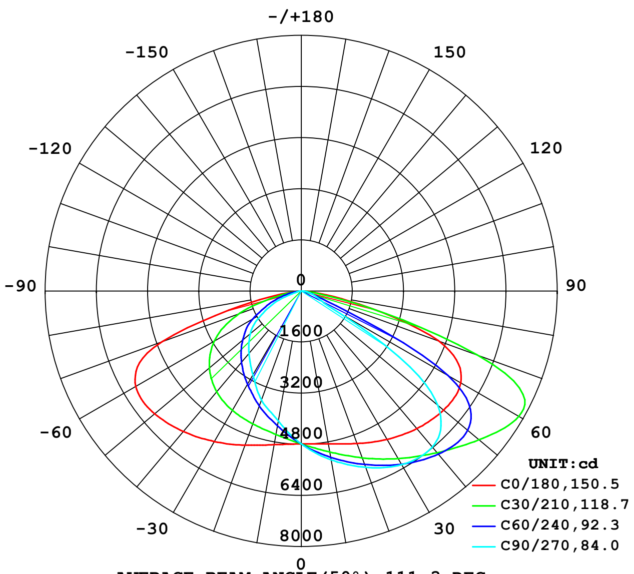
γ Range: 0 - 90DEG
γ Interval: 0.5DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

STREETLIGHT PHOTOMETRIC TEST REPORT

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		

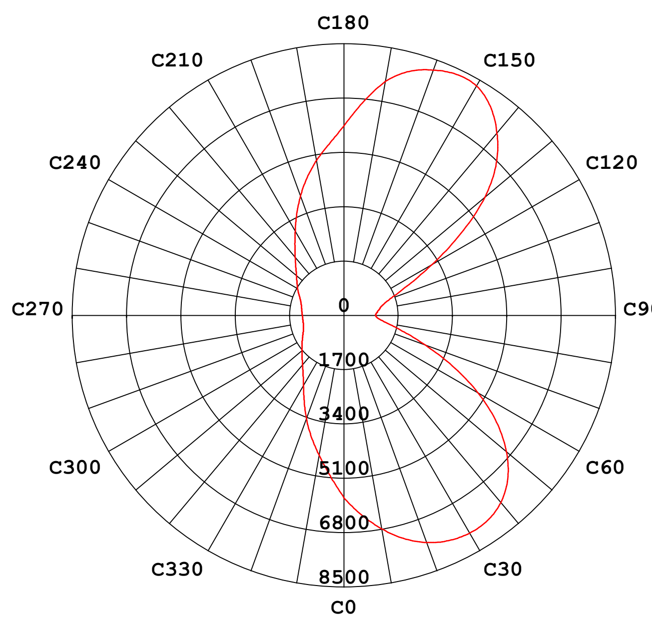
DATA OF LAMP		PHOTOMETRIC DATA		Eff: 160.31 lm/W	
MODEL	STREET LIGHT	Imax(cd)	8297	η street_up(%)	0.0
NOMINAL POWER(W)	127	LOR(%)	100.0	η street_down(%)	63.8
RATED VOLTAGE(V)	220	TOTAL FLUX(lm)	20021	η house_up(%)	0.0
NOMINAL FLUX(lm)	20021.1	MAXIMUM @(C,γ)	150,61.0	η house_down(%)	36.2
LAMPS INSIDE	1	η up(%)	0.0	76 FLASHAREA(m2)	0.05000
TEST VOLTAGE(V)	220	η down(%)	100.0	SLI	23.495

INTENSITY DISTRIBUTION DIAGRAM IN C PLANS



AVERAGE BEAM ANGLE(50%):111.3 DEG

MAX INTENSITY CONE SURFACE DISTRIBUTION DIAGRAM



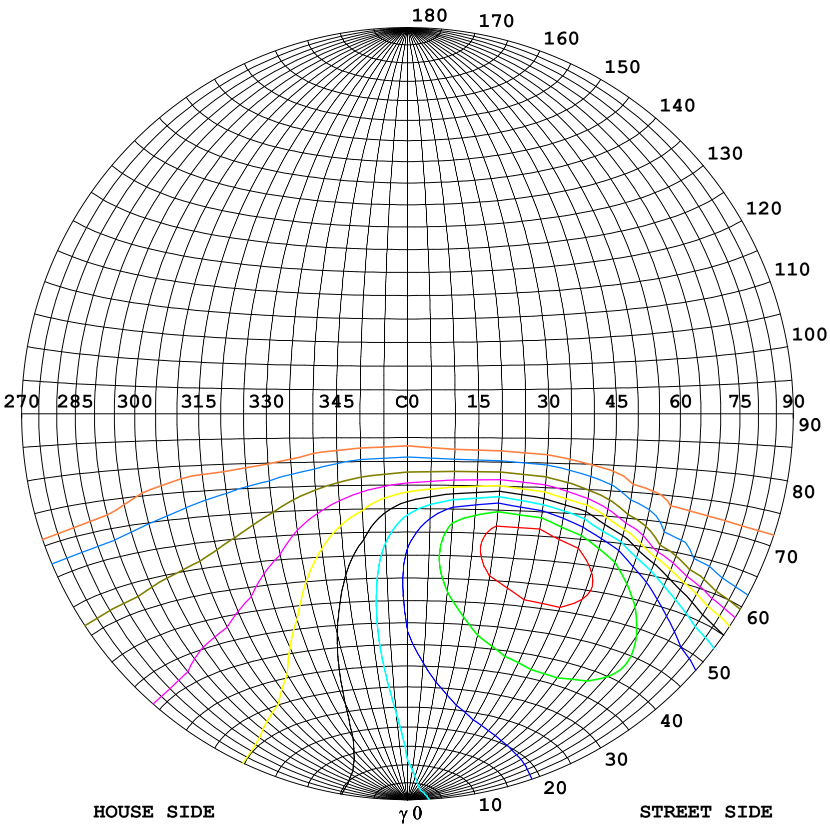
HOUSE SIDE Max @ γ = 61.0DEG STREET SIDE

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

STREETLIGHT ISOCANDELA DIAGRAM

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		



Classification:

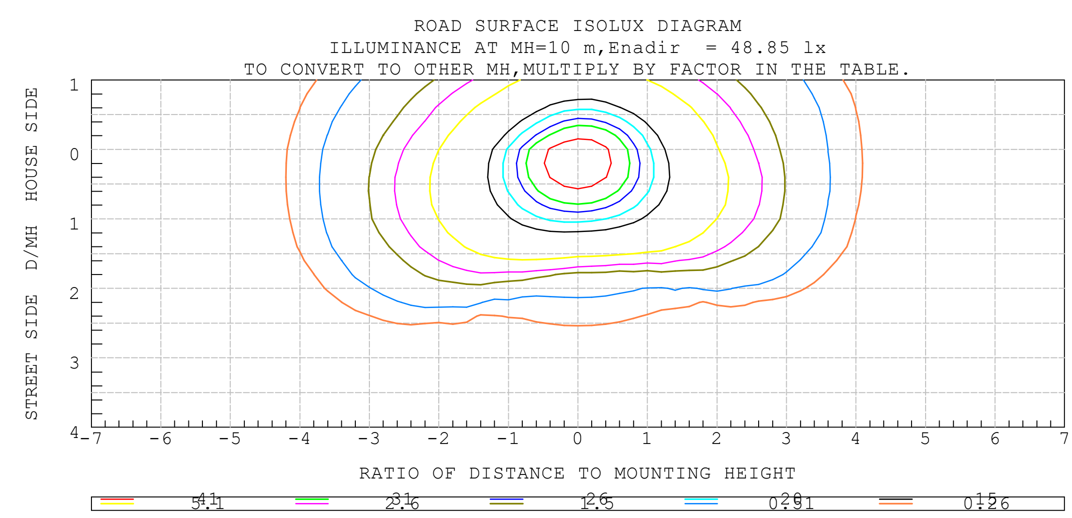
IES:Type II - Short
CIE:Average - Intermediate
IES:Cut-off
CIE:Semi-cut-off
Max.At80:50.97cd/klm
Max.At90:2.369cd/klm
Max.80-90:50.97cd/klm

ISOCANDELA DIAGRAM	
UNIT	cd
Imax=100%	8297
90%	7467
80%	6637
70%	5808
60%	4978
50%	4148
40%	3319
30%	2489
20%	1659
10%	830
5%	415

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

COEFFICIENT OF UTILIZATION CURVE AND ISOLUX DIAGRAM



HEIGHT	FACTOR
5 m	4.000
6 m	2.778
7 m	2.041
8 m	1.563
9 m	1.235
10 m	1.000
11 m	0.826
12 m	0.694
13 m	0.592
14 m	0.510
15 m	0.444
16 m	0.391

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ZONAL FLUX DIAGRAM

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	**TYPE:** STREET LIGHT	
SPEC.:	**DIM.:** 420X320X27	
MFR.: NASSLI		

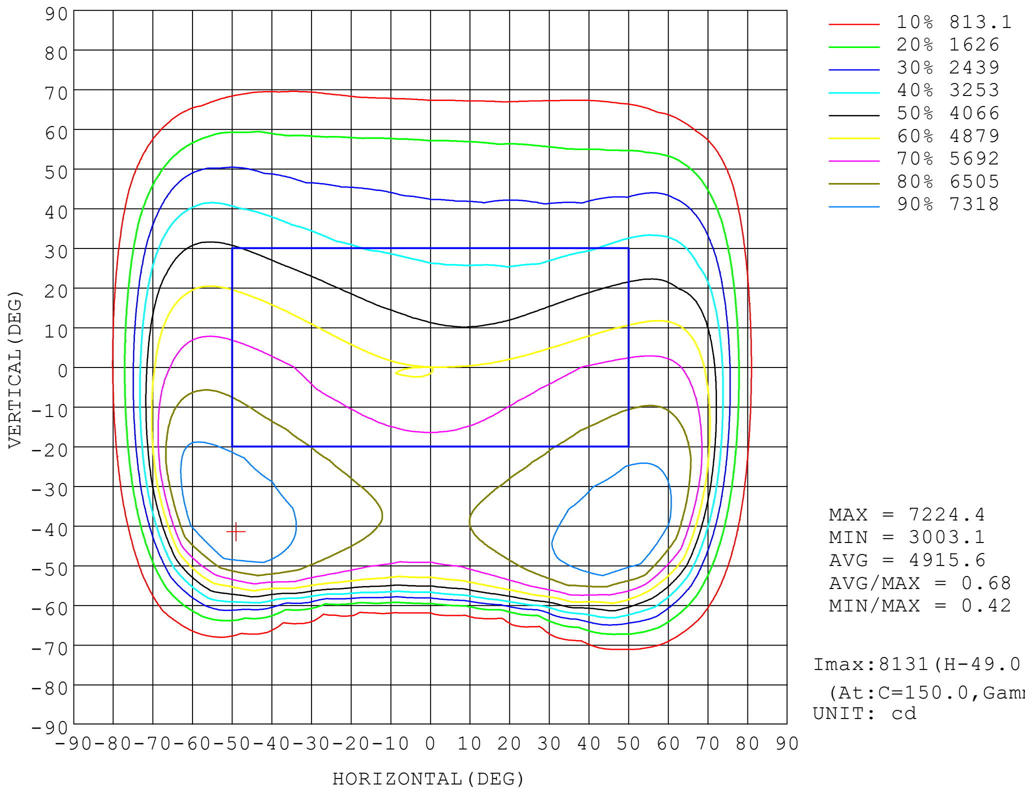
γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	Φ zone	Φ total	%lum,lamp
10	4856	5205	5364	5287	4883	4419	4141	4312	0- 10	457.9	457.9	2.29,2.29
20	5080	5684	5870	5826	5136	4113	3636	3904	10- 20	1376	1834	9.16,9.16
30	5362	6200	6277	6380	5439	3784	3031	3486	20- 30	2293	4128	20.6,20.6
40	5621	6733	6372	6931	5741	3389	2546	3049	30- 40	3162	7290	36.4,36.4
50	5804	7250	5563	7373	5979	2886	2087	2553	40- 50	3890	11180	55.8,55.8
60	5752	7270	1403	6873	5972	2239	1383	1948	50- 60	4160	15340	76.6,76.6
70	4558	3208	471.4	1759	4622	1298	538.7	1091	60- 70	3228	18568	92.7,92.7
80	1020	300.3	170.6	212.8	828.3	243.8	84.60	247.2	70- 80	1309	19877	99.3,99.3
90	45.05	33.22	36.91	39.52	39.26	46.79	35.81	36.47	80- 90	144.2	20021	100,100
100									90-100			
110									100-110			
120									110-120			
130									120-130			
140									130-140			
150									140-150			
160									150-160			
170									160-170			
180									170-180			
DEG	LUMINOUS INTENSITY:cd									UNIT:lm		

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ISOCANDELA DIAGRAM

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		



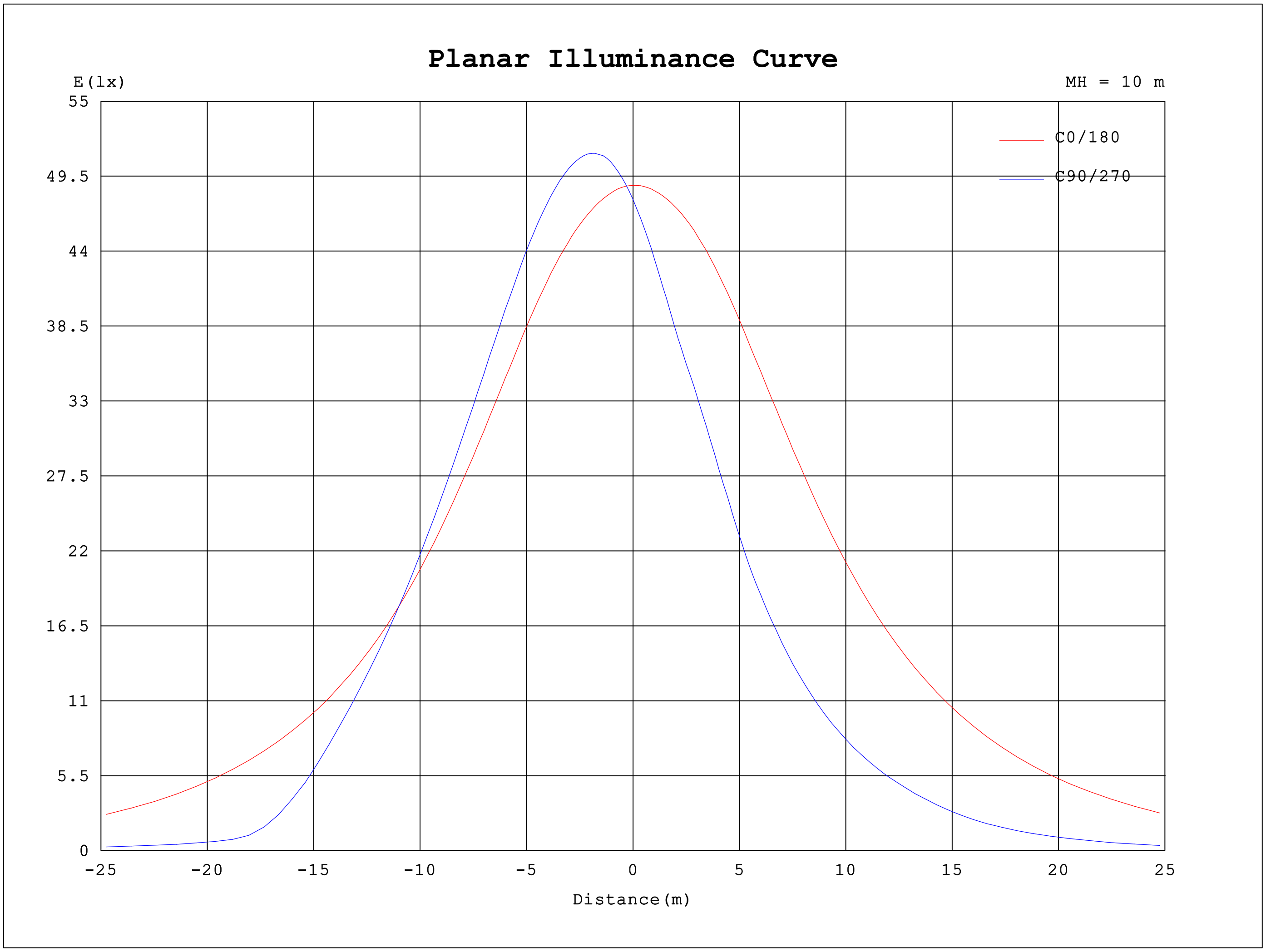
MAX = 7224.4
MIN = 3003.1
AVG = 4915.6
AVG/MAX = 0.68
MIN/MAX = 0.42

Imax:8131(H-49.0,V-41.5)
(At:C=150.0,Gamma=61.0)
UNIT: cd

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

Planar Illuminance Curve



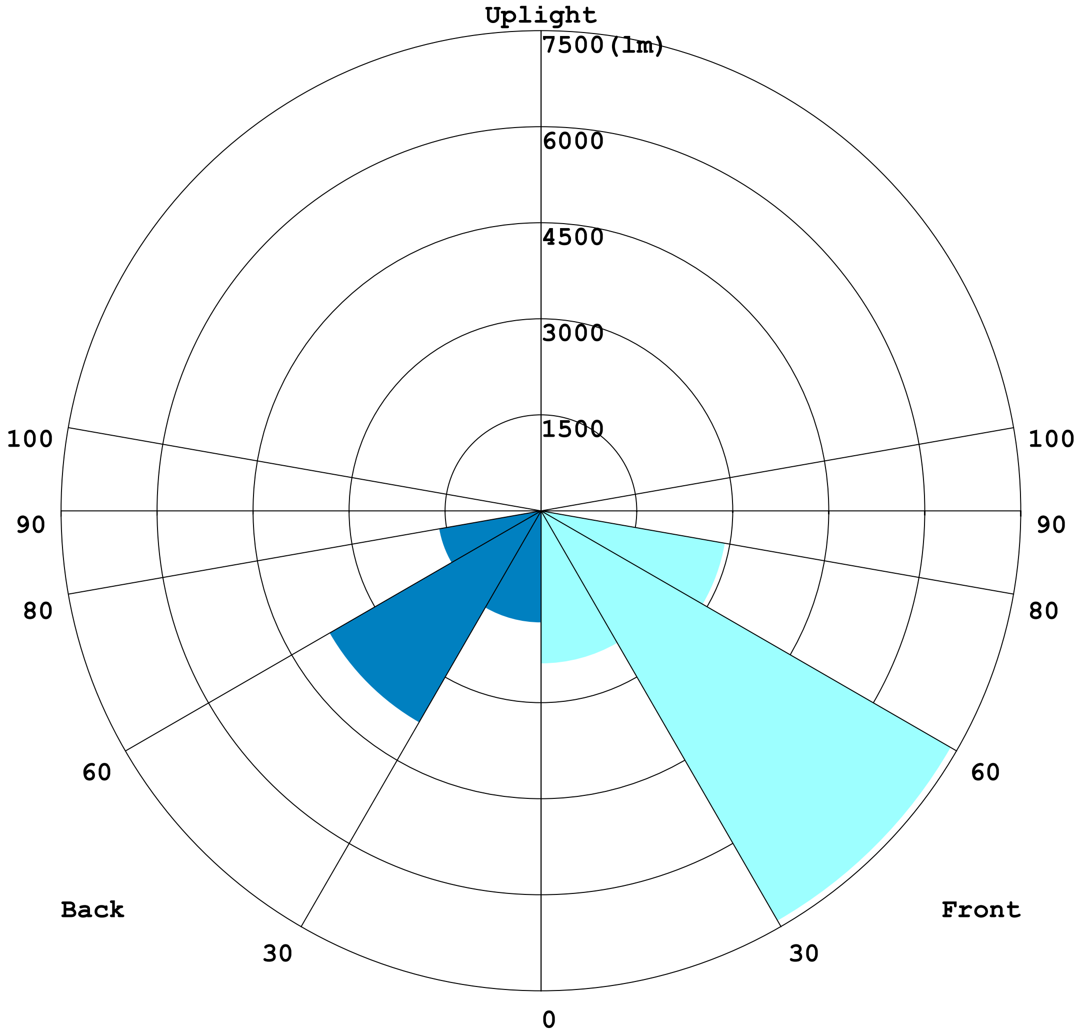
C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LCS REPORT

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		

LUMINAIRE CLASSIFICATION SYSTEM(LCS) GRAPH



C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

BUG REPORT

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		

IESNA Luminaire Flux Distribution Table:

Zone	Lumens	Luminaire %
FL - Front-Low(0-30)	2384.4	11.9
FM - Front-Medium(30-60)	7396.2	36.9
FH - Front-High(60-80)	2920.9	14.6
FVH - Front-Very High(80-90)	77.983	0.4
Total Forward Light	12780	63.8

BL - Back-Low(0-30)	1743.3	8.7
BM - Back-Medium(30-60)	3815.9	19.1
BH - Back-High(60-80)	1616.1	8.1
BVH - Back-Very High(80-90)	66.23	0.3
Total Back Light	7241.6	36.2

UL - Uplight-Low(90-100)	0	0.0
UH - Uplight-High(100-180)	0	0.0
Total Up Light	0	0.0

BUG(Back,Up,Glare) Rating	B3-U0-G3

Zone	Downward Lumens	Upward Lumens	Total Lumens
House Side	7241.6	0	7241.6
Street Side	12780	0	12780

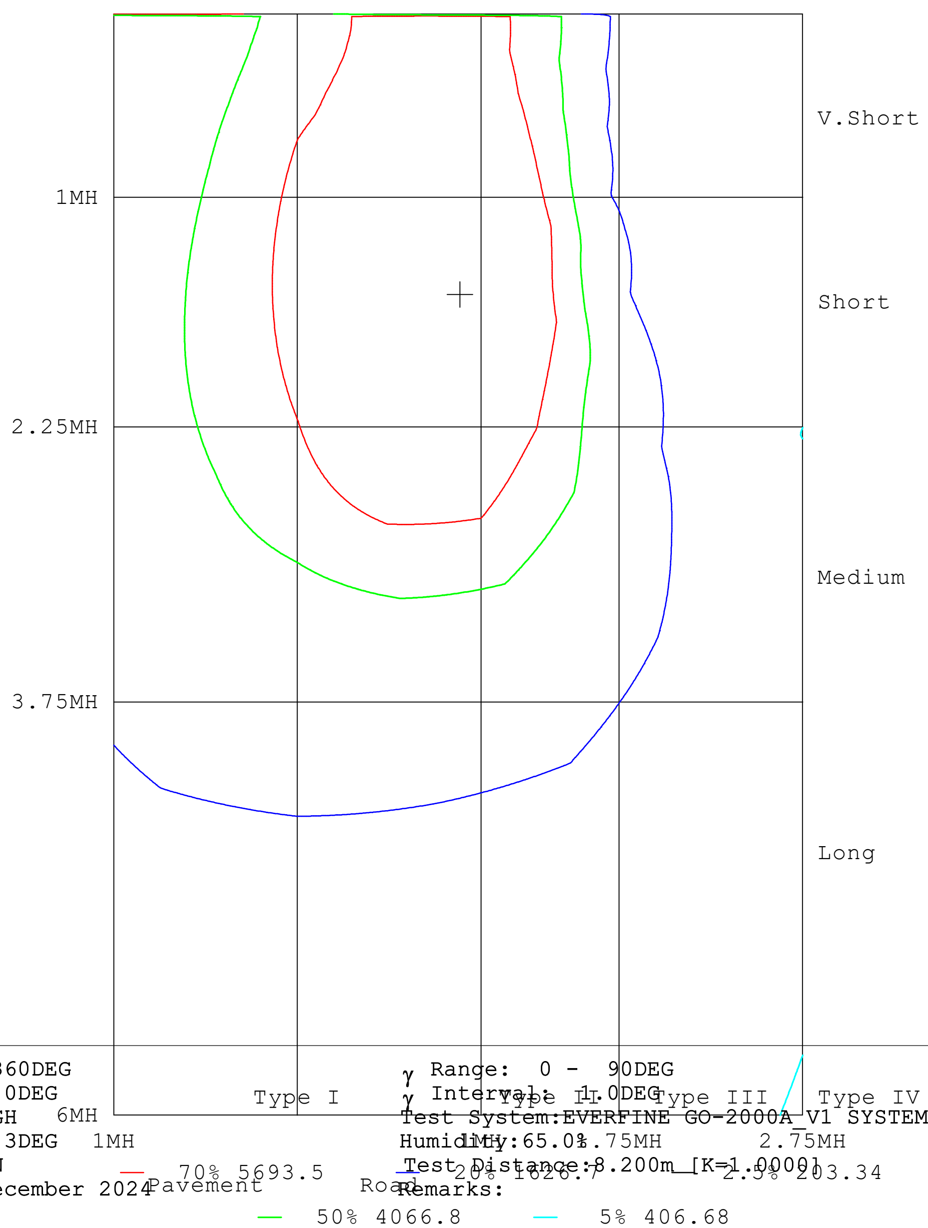
C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

ROAD ISOCANDELA REPORT

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm		
NAME: NSL-STL-127W-6500K-IP66	TYPE: STREET LIGHT	
SPEC.:	DIM.: 420X320X27	
MFR.: NASSLI		

ROAD SURFACE ISOCANDELA DIAGRAM



C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm**		
NAME: NSL-STL-127W-6500K-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

Table--1 **UNIT: cd**

C(DEG) / γ (DEG)	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180
0	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786
5	4801	4851	4896	4938	4973	5008	5038	5061	5078	5088	5089	5084	5069	5048	5019	4982	4939	4889	4813
10	4856	4955	5040	5115	5176	5234	5282	5321	5347	5364	5368	5363	5342	5311	5263	5201	5127	5032	4883
15	4952	5099	5219	5322	5402	5475	5530	5570	5601	5623	5628	5628	5611	5580	5525	5444	5344	5217	4997
20	5080	5271	5428	5555	5647	5722	5776	5816	5845	5870	5877	5883	5873	5851	5800	5713	5588	5424	5136
25	5221	5460	5651	5802	5902	5977	6021	6051	6074	6099	6109	6124	6128	6122	6081	5992	5847	5644	5288
30	5362	5649	5882	6057	6164	6236	6259	6257	6255	6277	6292	6333	6368	6392	6368	6280	6110	5869	5439
35	5497	5842	6118	6321	6439	6495	6476	6429	6378	6379	6406	6489	6577	6655	6667	6582	6393	6102	5596
40	5621	6038	6372	6606	6724	6742	6665	6541	6409	6372	6409	6558	6735	6897	6965	6908	6698	6345	5741
45	5726	6227	6639	6914	7015	6976	6795	6536	6267	6157	6212	6463	6784	7092	7259	7256	7029	6601	5873
50	5804	6424	6941	7248	7321	7179	6787	6272	5787	5563	5631	6046	6610	7194	7552	7624	7401	6874	5979
55	5835	6625	7271	7598	7610	7253	6455	5514	4609	4141	4228	4967	5973	7038	7776	8003	7792	7161	6036
60	5752	6766	7538	7849	7699	6838	5290	3395	1947	1403	1423	2208	3906	6140	7607	8290	8126	7394	5972
65	5423	6686	7498	7645	6864	5118	2044	749	659	650	660	701	898	2998	6050	7783	8068	7404	5650
70	4558	5839	6379	5838	4446	1966	587	490	472	471	476	496	532	797	2724	5212	6550	6132	4622
75	2625	3032	3289	2749	1571	571	342	323	311	309	310	324	341	383	782	1984	2802	2745	2355
80	1020	926	868	697	397	203	177	172	170	171	170	175	176	190	236	464	703	697	828
85	211	146	141	126	89.9	72.2	71.3	70.4	71.6	69.3	71.9	72.8	73.9	76.1	77.5	97.7	147	102	140
90	45.1	40.4	36.5	34.0	33.2	33.2	33.7	34.7	36.1	36.9	38.6	39.5	39.8	39.8	39.3	38.8	38.6	38.6	39.3

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.18V I:0.5817A P:124.89W PF:0.9750 Freq:59.98Hz **UTHDi:0.00% ITHDi:0.00% KDisp:0 Lamp Flux:20021.1x1 lm**		
NAME: NSL-STL-127W-6500K-IP66	**TYPE: STREET LIGHT**	
SPEC.:	**DIM.: 420X320X27**	
MFR.: NASSLI		

Table--2 **UNIT: cd**

C(DEG) / γ (DEG)	190	200	210	220	230	240	250	260	270	280	290	300	310	320	330	340	350		
0	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786	4786		
5	4761	4712	4664	4621	4576	4538	4504	4481	4468	4466	4475	4495	4525	4565	4612	4665	4722		
10	4775	4666	4562	4468	4371	4288	4216	4168	4141	4139	4160	4203	4268	4355	4456	4568	4689		
15	4831	4656	4487	4333	4182	4060	3961	3900	3882	3886	3905	3946	4035	4160	4317	4498	4695		
20	4910	4668	4429	4217	4009	3839	3716	3668	3636	3639	3672	3730	3820	3988	4200	4456	4735		
25	5002	4689	4377	4097	3815	3595	3476	3396	3336	3337	3395	3488	3591	3801	4090	4431	4791		
30	5093	4709	4314	3958	3610	3369	3241	3114	3031	3035	3104	3242	3369	3603	3970	4406	4854		
35	5178	4714	4236	3798	3399	3143	3008	2854	2782	2786	2846	3012	3144	3402	3831	4365	4901		
40	5243	4692	4119	3606	3171	2910	2764	2608	2546	2552	2601	2775	2911	3188	3667	4296	4926		
45	5280	4631	3959	3383	2923	2662	2506	2372	2320	2326	2373	2527	2666	2958	3474	4187	4917		
50	5277	4521	3754	3121	2650	2394	2242	2139	2087	2094	2139	2261	2407	2700	3235	4031	4860		
55	5205	4346	3487	2816	2350	2114	1972	1857	1792	1802	1865	1997	2129	2415	2954	3819	4745		
60	5002	4060	3141	2460	2017	1763	1550	1437	1383	1395	1449	1611	1803	2095	2616	3518	4522		
65	4559	3592	2676	2031	1579	1306	1123	1035	1003	997	1068	1184	1358	1710	2189	3076	4086		
70	3684	2816	2045	1469	1126	938	759	625	539	582	668	818	980	1202	1674	2423	3323		
75	2151	1675	1169	912	715	579	363	177	172	176	203	450	610	776	996	1529	2171		
80	816	635	488	282	205	253	112	85.7	84.6	82.0	85.6	152	268	226	395	636	932		
85	153	94.1	77.7	72.6	65.3	60.9	50.7	45.5	41.9	39.6	42.1	47.9	44.1	53.0	69.7	119	196		
90	43.4	46.2	47.4	47.4	46.1	43.7	40.8	38.1	35.8	34.4	34.0	34.5	35.7	37.2	39.2	40.6	41.0		

C Range: 0 - 360DEG
C Interval: 10.0DEG
Test Speed: HIGH
Temperature:25.3DEG
Operators:MOMEN
Test Date:30 December 2024

γ Range: 0 - 90DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-2000A_V1 SYSTEM V2.00.454
Humidity:65.0%
Test Distance:8.200m [K=1.0000]
Remarks:

INVENTRONICS

Features

- Compact Metal Case with Excellent Thermal Performance
- Full Power at Wide Output Current Range (Constant Power)
- Adjustable Output Current (AOC) with Programmability
- Isolated 1-5V/1-10V/10V PWM/3-Timer-Modes Dimmable
- Output Lumen Compensation
- Input Surge Protection: DM 6kV, CM 10kV
- All-Around Protection: OVP, SCP, OTP
- IP66 / IP67 and UL Dry / Damp / Wet Location
- SELV Output
- TYPE HL, for use in a Class I, Division 2 hazardous (Classified) location
- 5 Years Warranty

Description

The EUM-150SxxxDx series is a 150W, constant-current, programmable and IP66/IP67 rated LED driver that operates from 90-305Vac input with excellent power factor. It is created for many lighting applications including high bay, high mast and roadway, etc. The high efficiency of these drivers and compact metal case enables them to run cooler, significantly improving reliability and extending product life. To ensure trouble-free operation, protection is provided against input surge, output over voltage, short circuit, and over temperature.

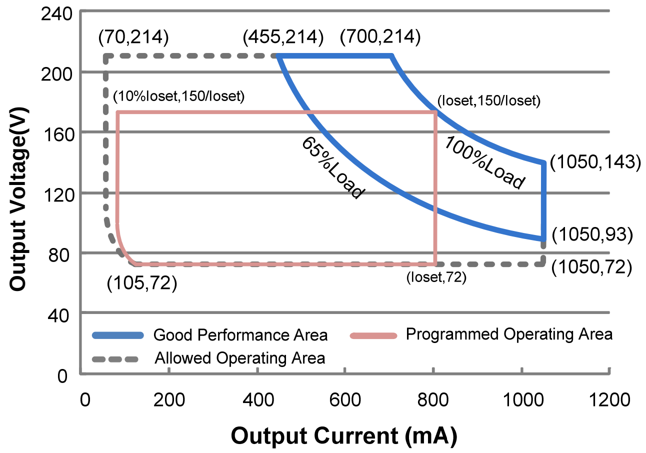
Models

Adjustable Output Current Range	Full-Power Current Range (1)	Default Output Current	Input Voltage Range(2)	Output Voltage Range	Max. Output Power	Typical Efficiency (3)	Typical Power Factor		Model Number (5)
							120Vac	220Vac	
70-1050mA	700-1050mA	700mA	90~305 Vac/ 127~300 Vdc	72~214 Vdc	150W	93.0%	0.99	0.96	EUM-150S105Dx
105-1500mA	1050-1500mA	1050mA	90~305 Vac/ 127~300 Vdc	50~143 Vdc	150W	93.5%	0.99	0.96	EUM-150S150Dx
140-2100mA	1400-2100mA	1400mA	90~305 Vac/ 127~300 Vdc	36~107 Vdc	150W	92.0%	0.99	0.96	EUM-150S210Dx(4)
280-4200mA	2800-4200mA	3150mA	90~305 Vac/ 127~300 Vdc	18 ~ 54 Vdc	150W	91.5%	0.99	0.96	EUM-150S420Dx(4)

Notes: (1) Output current range with constant power at 150W
(2) Certified input voltage range: UL, FCC 100-277Vac; otherwise 100-240Vac.
(3) Measured at 100% load and 220Vac input (see below "General Specifications" for details).
(4) SELV output.
(5) x = G are UL Recognized, ENEC and CCC, etc. models; x = T are UL Class P models; x = B are BIS models.

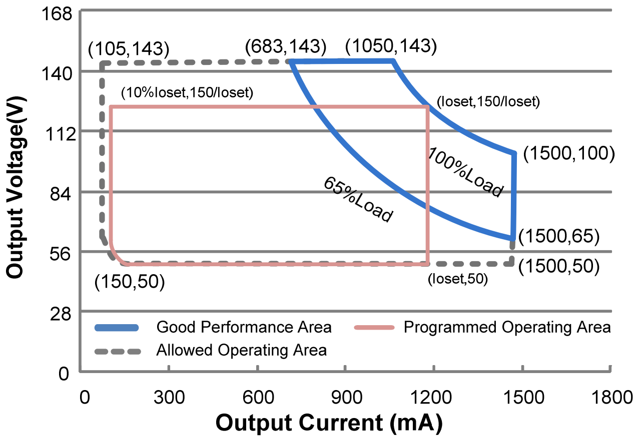
I-V Operation Area

EUM-150S105Dx



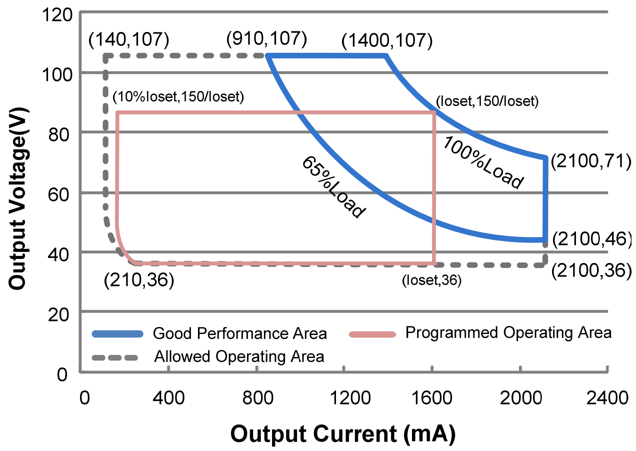
Note: 700mA≤Ioset≤1050mA

EUM-150S150Dx



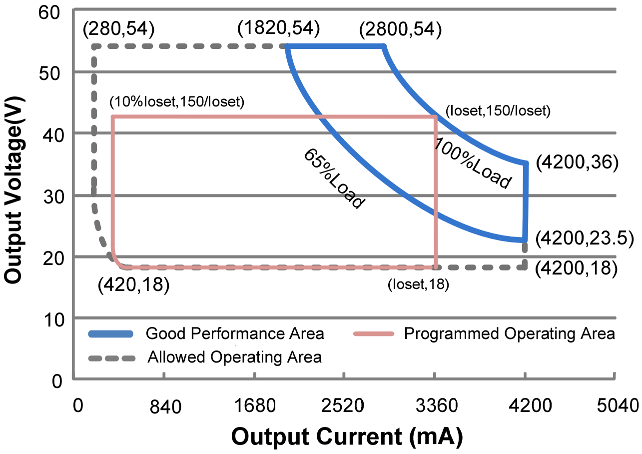
Note: 1050mA≤Ioset≤1500mA

EUM-150S210Dx



Note: 1400mA≤Ioset≤2100mA

EUM-150S420Dx



Note: 2800mA≤Ioset≤4200mA

Input Specifications

Parameter	Min.	Typ.	Max.	Notes
Input AC Voltage	90 Vac	-	305 Vac	
Input DC Voltage	127 Vdc	-	300 Vdc	
Input Frequency	47 Hz	-	63 Hz	
Leakage Current	-	-	0.75 MIU	UL8750; 277Vac/ 60Hz
	-	-	0.70 mA	IEC60598-1; 240Vac/ 60Hz,
Input AC Current	-	-	1.50 A	Measured at 100% load and 120 Vac input.
	-	-	0.80 A	Measured at 100% load and 220 Vac input.

Input Specifications (Continued)

Parameter	Min.	Typ.	Max.	Notes
Inrush Current(I^2t)	-	-	3.55 A^2s	At 220Vac input, 25°C cold start, duration=220 μs, 10%Ipk-10%Ipk. See Inrush Current Waveform for the details.
PF	0.9	-	-	At 100-277Vac, 50-60Hz, 65%-100% Load (97.5-150W)
THD	-	-	20%	
THD	-	-	10%	At 220-240Vac, 50-60Hz, 75%-100% Load (112.5-150W)

Output Specifications

Parameter	Min.	Typ.	Max.	Notes
Output Current Tolerance	-5%Ioset	-	5%Ioset	At 100% load condition
Output Current Setting(Ioset) Range EUM-150S105Dx EUM-150S150Dx EUM-150S210Dx EUM-150S420Dx	 70 mA 105 mA 140 mA 280 mA	 - - - -	 1050 mA 1500 mA 2100 mA 4200 mA	
Output Current Setting Range with Constant Power EUM-150S105Dx EUM-150S150Dx EUM-150S210Dx EUM-150S420Dx	 700 mA 1050 mA 1400 mA 2800 mA	 - - - -	 1050 mA 1500 mA 2100 mA 4200 mA	
Total Output Current Ripple (pk-pk)	-	5%Iomax	10%Iomax	At 100% load condition. 20 MHz BW
Output Current Ripple at < 200 Hz (pk-pk)	-	2%Iomax	-	At 100% load condition. Only this component of ripple is associated with visible flicker.
Startup Overshoot Current	-	-	10%Iomax	At 100% load condition
No Load Output Voltage EUM-150S105Dx EUM-150S150Dx EUM-150S210Dx EUM-150S420Dx	 - - - -	 - - - -	 240 V 160 V 120 V 60 V	
Line Regulation	-	-	±0.5%	Measured at 100% load
Load Regulation	-	-	±1.5%	
Turn-on Delay Time	-	-	0.5 s	Measured at 120-277Vac input, 65%-100% Load
Temperature Coefficient of Ioset	-	0.03%/°C	-	Case temperature = 0°C ~Tc max

General Specifications

Parameter	Min.	Typ.	Max.	Notes
Efficiency at 120 Vac input: EUM-150S105Dx Io= 700 mA Io=1050 mA EUM-150S150Dx Io=1050 mA Io=1500 mA EUM-150S210Dx Io=1400 mA Io=2100 mA EUM-150S420Dx Io=2800 mA Io=4200 mA	 88.5% 89.0% 89.0% 89.5% 87.5% 88.0% 87.0% 86.5%	 90.5% 91.0% 91.0% 91.5% 89.5% 90.0% 89.0% 88.5%	 - - - - - - - -	Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
Efficiency at 220 Vac input: EUM-150S105Dx Io= 700 mA Io=1050 mA EUM-150S150Dx Io=1050 mA Io=1500 mA EUM-150S210Dx Io=1400 mA Io=2100 mA EUM-150S420Dx Io=2800 mA Io=4200 mA	 90.5% 91.0% 91.0% 91.5% 89.5% 90.0% 89.5% 89.0%	 92.5% 93.0% 93.0% 93.5% 91.5% 92.0% 91.5% 91.0%	 - - - - - - - -	Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
Efficiency at 277 Vac input: EUM-150S105Dx Io= 700 mA Io=1050 mA EUM-150S150Dx Io=1050 mA Io=1500 mA EUM-150S210Dx Io=1400 mA Io=2100 mA EUM-150S420Dx Io=2800 mA Io=4200 mA	 91.0% 91.5% 91.5% 91.5% 90.0% 90.0% 89.5% 89.0%	 93.0% 93.5% 93.5% 93.5% 92.0% 92.0% 91.5% 91.0%	 - - - - - - - -	Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
MTBF	-	333,000 Hours	-	Measured at 220Vac input, 80%Load and 25°C ambient temperature (MIL-HDBK-217F)
Lifetime	-	106,000 Hours	-	Measured at 220Vac input, 80%Load and 70°C case temperature; See lifetime vs. Tc curve for the details
Operating Case Temperature for Safety Tc_s	-40°C	-	+90°C	
Operating Case Temperature for Warranty Tc_w	-40°C	-	+80°C	Case temperature for 5 years warranty Humidity: 10% RH to 95% RH;
Storage Temperature	-40°C	-	+85°C	Humidity: 5%RH to 95%RH
Dimensions Inches (L × W × H) Millimeters (L × W × H)	 6.34 × 2.36 ×1.44 161 × 60 × 36.5			With mounting ear 7.01 × 2.36 ×1.44 178 × 60 × 36.5
Net Weight	-	735 g	-	

Dimming Specifications

Parameter		Min.	Typ.	Max.	Notes
Absolute Maximum Voltage on the Vdim (+) Pin		-20 V	-	20 V	
Source Current on Vdim (+)Pin		200 μA	300 μA	450 μA	Vdim(+) = 0 V
Dimming Output Range	EUM-150S105Dx EUM-150S150Dx EUM-150S210Dx EUM-150S420Dx	10%Ioset	-	Ioset	700 mA ⩽ Ioset ⩽ 1050 mA 1050 mA ⩽ Ioset ⩽ 1500 mA 1400 mA ⩽ Ioset ⩽ 2100 mA 2800 mA ⩽ Ioset ⩽ 4200 mA
	EUM-150S105Dx EUM-150S150Dx EUM-150S210Dx EUM-150S420Dx	70 mA 105 mA 140 mA 280 mA	-	Ioset	70 mA ⩽ Ioset ＜ 700 mA 105 mA ⩽ Ioset ＜ 1050 mA 140 mA ⩽ Ioset ＜ 1400 mA 280 mA ⩽ Ioset ＜ 2800 mA
Recommended Dimming Range for 1-5V		0.25 V	-	4.75 V	Dimming mode set to 1-5V in PC interface.
Recommended Dimming Range for 1-10V		1 V	-	9 V	Default 1-10V dimming mode with positive logic.
PWM_in High Level		-	10V	-	
PWM_in Low Level		-	0V	-	
PWM_in Frequency Range		200 Hz	-	2 KHz	
PWM_in Duty Cycle		0%	-	100%	

Safety &EMC Compliance

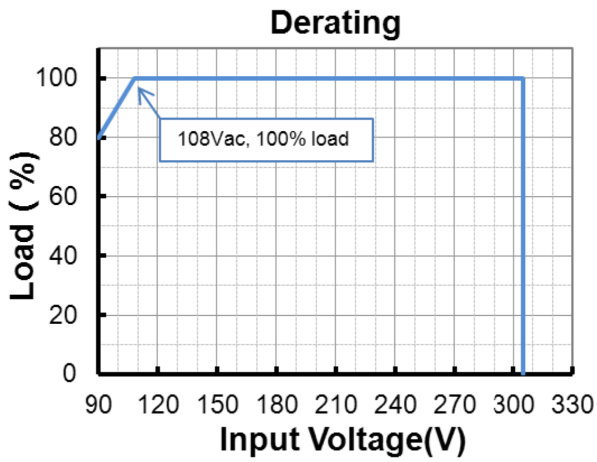
Safety Category	Standard
UL/CUL	UL8750,CAN/CSA-C22.2 No. 250.13
ENEC & CE	EN 61347-1, EN 61347-2-13
CB	IEC 61347-1, IEC 61347-2-13
CCC	GB 19510.1, GB 19510.14
PSE	J 61347-1, J 61347-2-13
KS	KS C 7655
BIS	IS 15885(Part2/Sec13)
EAC	ГОСТ Р МЭК 61347-1, ГОСТ IEC 61347-2-13
NOM	NOM-058-SCFI
EMI Standards	**Notes**
EN 55015/GB 17743/KN 15(1)	Conducted emission Test &Radiated emission Test
EN 61000-3-2/GB 17625.1	Harmonic current emissions
EN 61000-3-3	Voltage fluctuations & flicker

Safety &EMC Compliance (Continued)

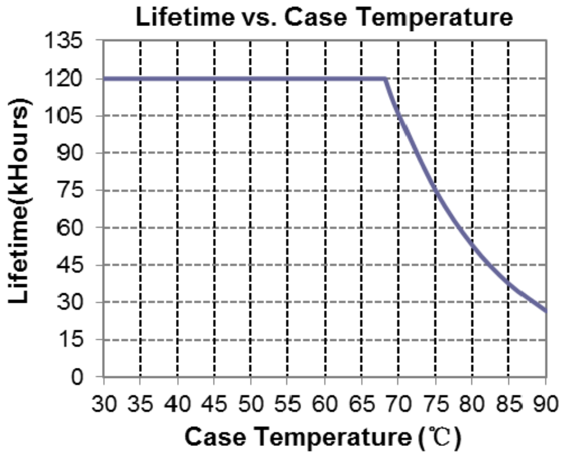
EMI Standards	Notes
FCC Part 15[1]	ANSI C63.4 Class B
	This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: [1] this device may not cause harmful interference, and [2] this device must accept any interference received, including interference that may cause undesired Operation.
EMS Standards	**Notes**
EN 61000-4-2	Electrostatic Discharge (ESD): 8 kV air discharge, 4 kV contact discharge
EN 61000-4-3	Radio-Frequency Electromagnetic Field Susceptibility Test-RS
EN 61000-4-4	Electrical Fast Transient / Burst-EFT
EN 61000-4-5	Surge Immunity Test: AC Power Line: Differential Mode 6 kV, Common Mode 10 kV
EN 61000-4-6	Conducted Radio Frequency Disturbances Test-CS
EN 61000-4-8	Power Frequency Magnetic Field Test
EN 61000-4-11	Voltage Dips
EN 61547	Electromagnetic Immunity Requirements Applies To Lighting Equipment

Note: (1) This LED driver meets the EMI specifications above, but EMI performance of a luminaire that contains it depends also on the other devices connected to the driver and on the fixture itself.

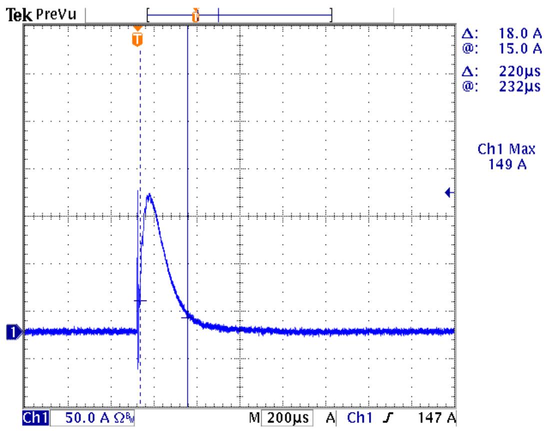
Derating



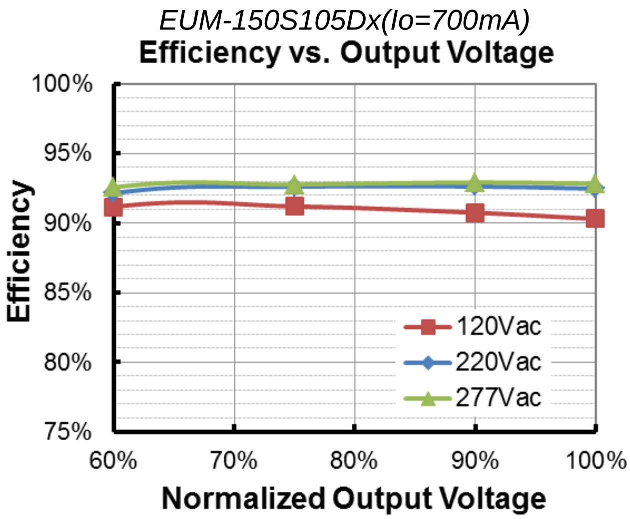
Lifetime vs. Case Temperature

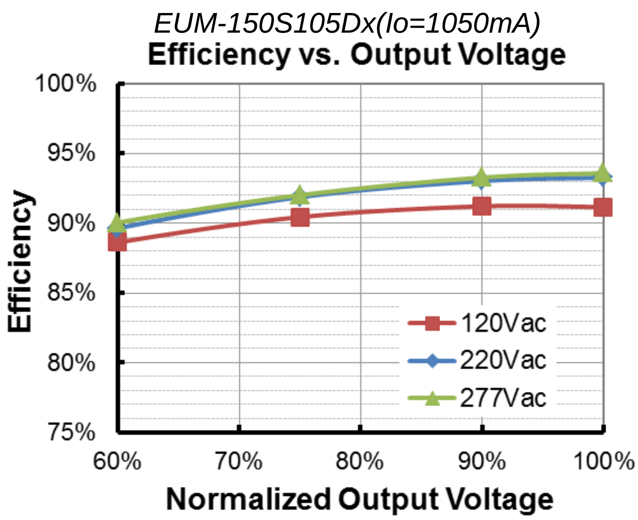


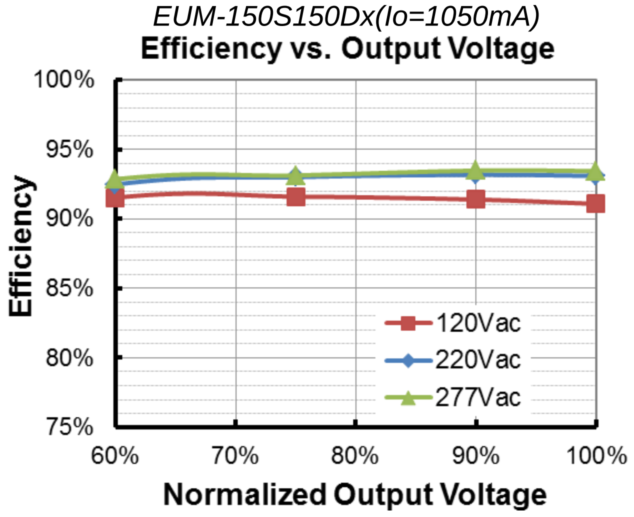
Inrush Current Waveform

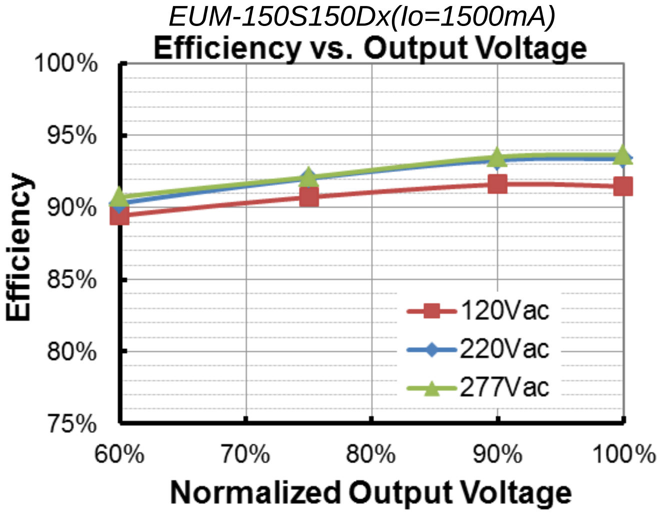


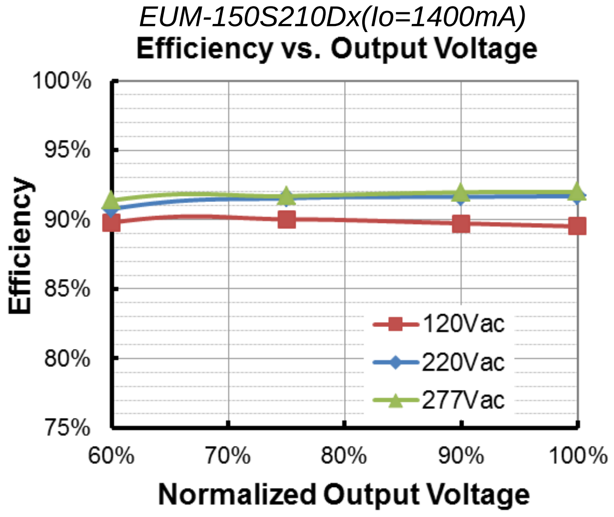
Efficiency vs. Load

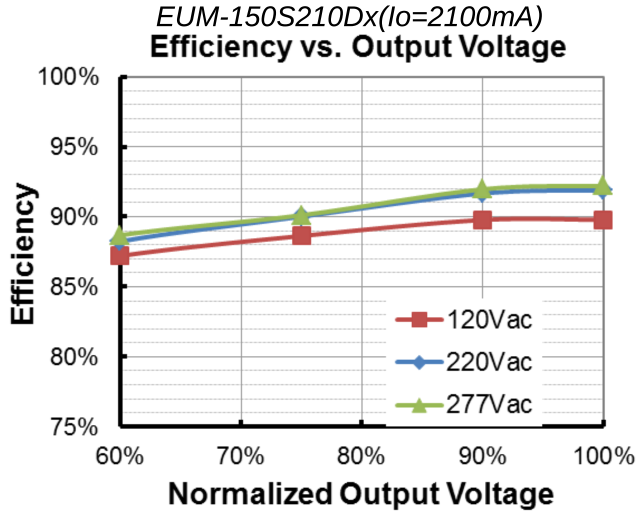


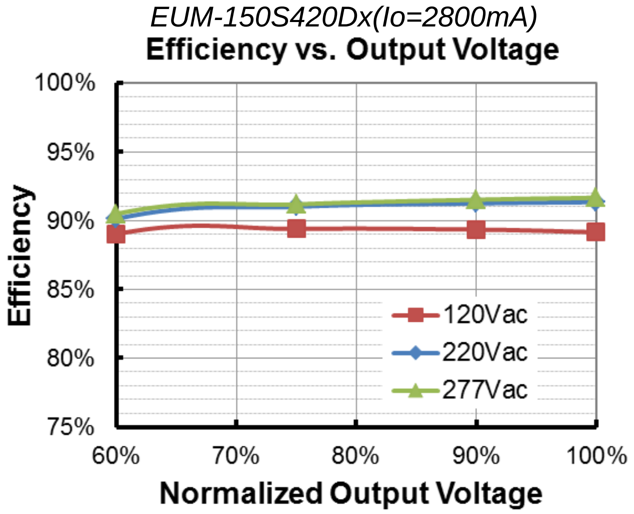


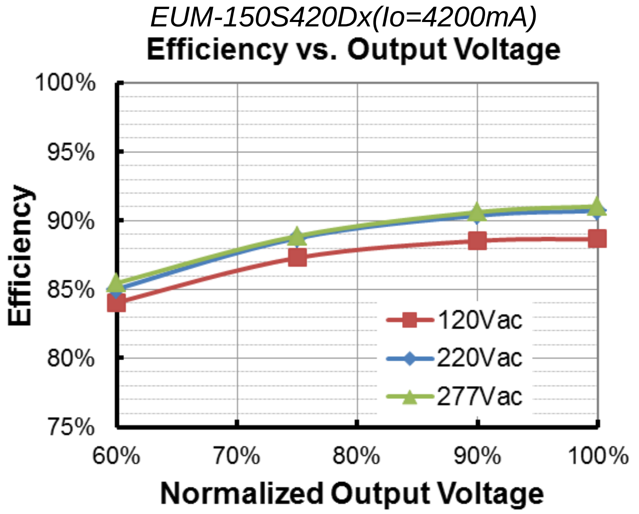

EUM-150S150Dx(Io=1050mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac


EUM-150S150Dx(Io=1500mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac

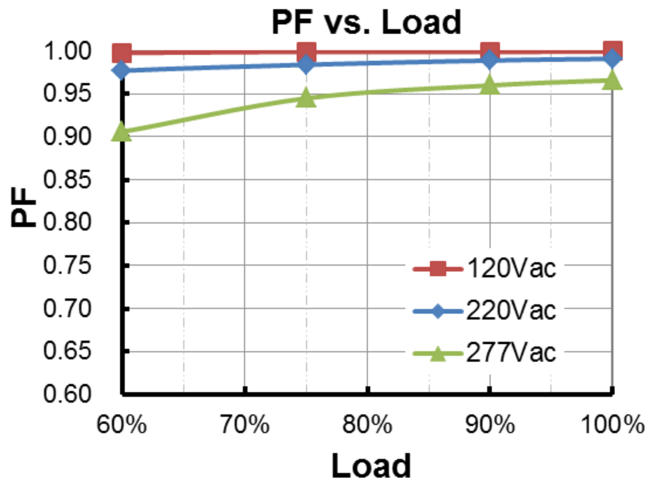

EUM-150S210Dx(Io=1400mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac


EUM-150S210Dx(Io=2100mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac

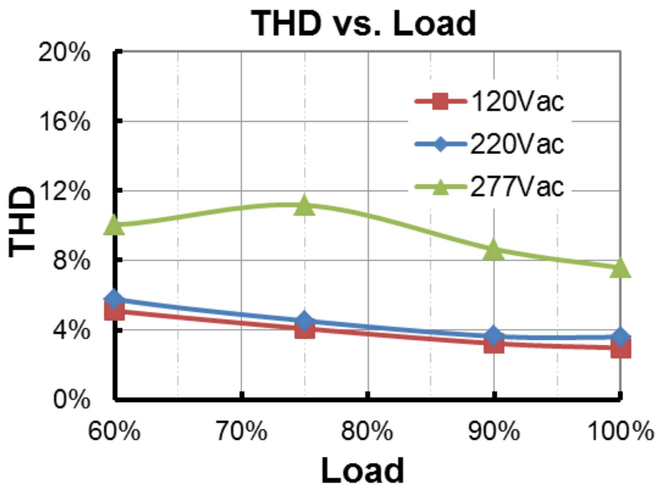

EUM-150S420Dx(Io=2800mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac


EUM-150S420Dx(Io=4200mA)
Efficiency vs. Output Voltage
100%
95%
90%
85%
80%
75%
Efficiency
60%
70%
80%
90%
100%
Normalized Output Voltage
120Vac
220Vac
277Vac

Power Factor



Total Harmonic Distortion



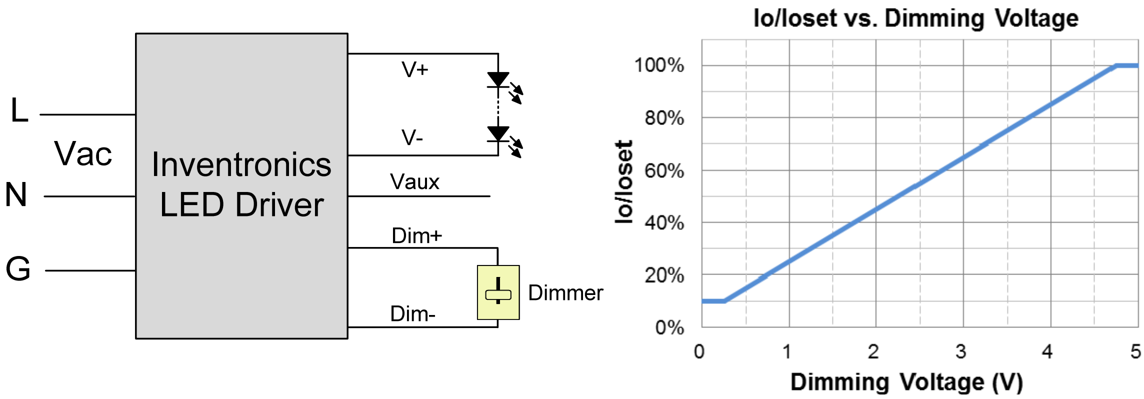
Protection Functions

Parameter	Notes
Over Temperature Protection	Decreases output current, returning to normal after over temperature is removed.
Short Circuit Protection	Auto Recovery. No damage will occur when any output is short circuited. The output shall return to normal when the fault condition is removed.
Over Voltage Protection	Limits output voltage at no load and in case the normal voltage limit fails.

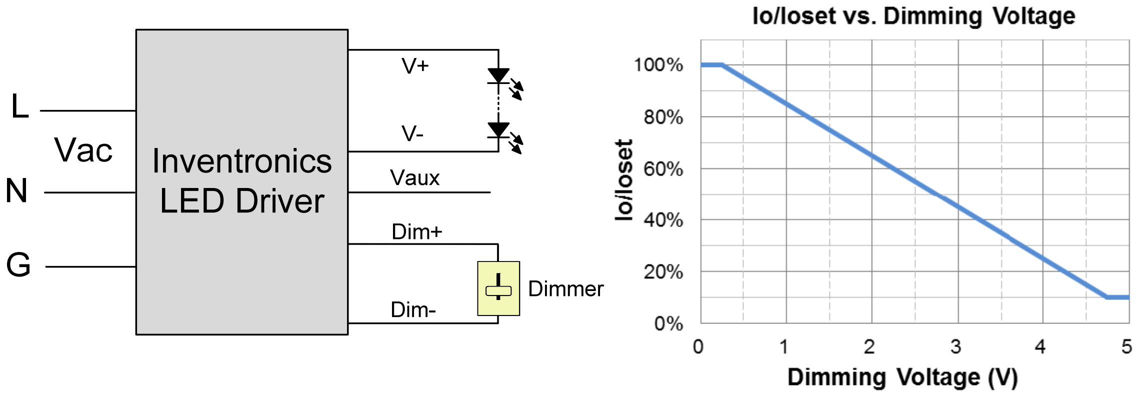
Dimming

- **1-5V Dimming**

The recommended implementation of the dimming control is provided below.



Implementation 1: Positive logic



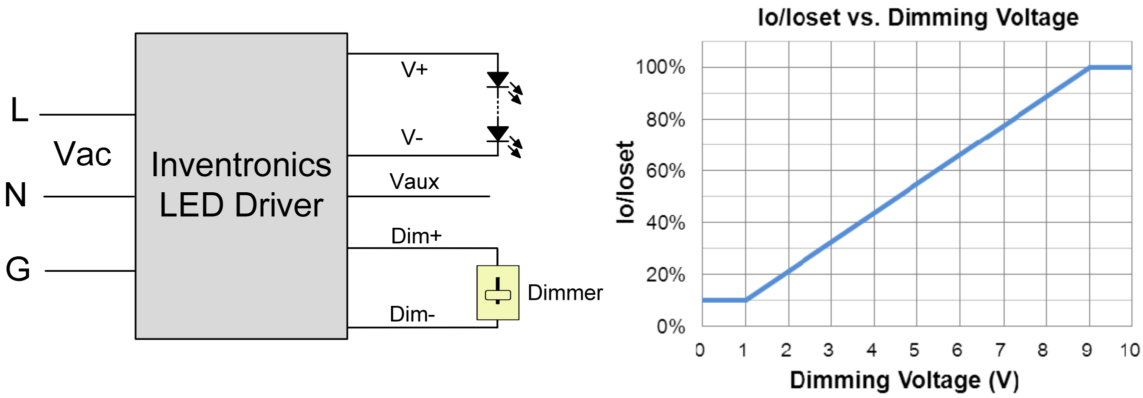
Implementation 2: Negative logic

Notes:

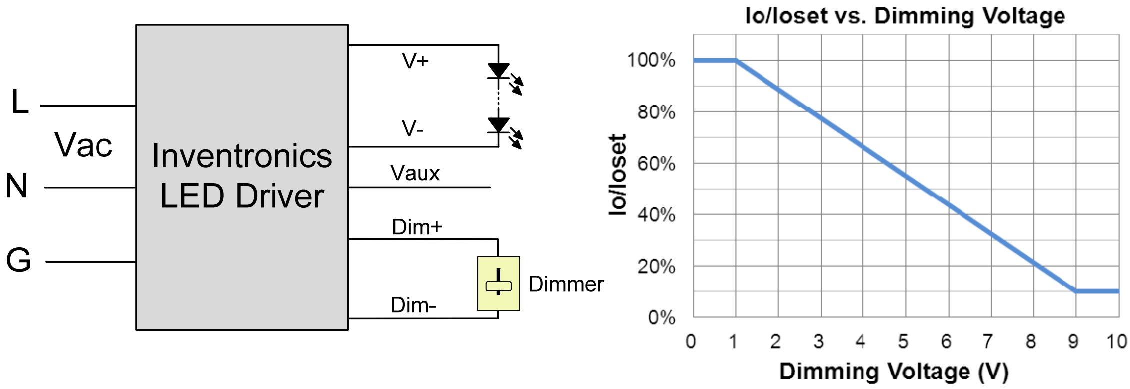
1. Do NOT connect Dim− to the output V− or V+, otherwise the driver will not work properly.
2. The dimmer can also be replaced by an active 1-5V voltage source signal or passive components like zener.
3. When 1-5V negative logic dimming mode and Dim+ is open, the driver will output maximum current.

- **1-10V Dimming**

The recommended implementation of the dimming control is provided below.



Implementation 3: Positive logic



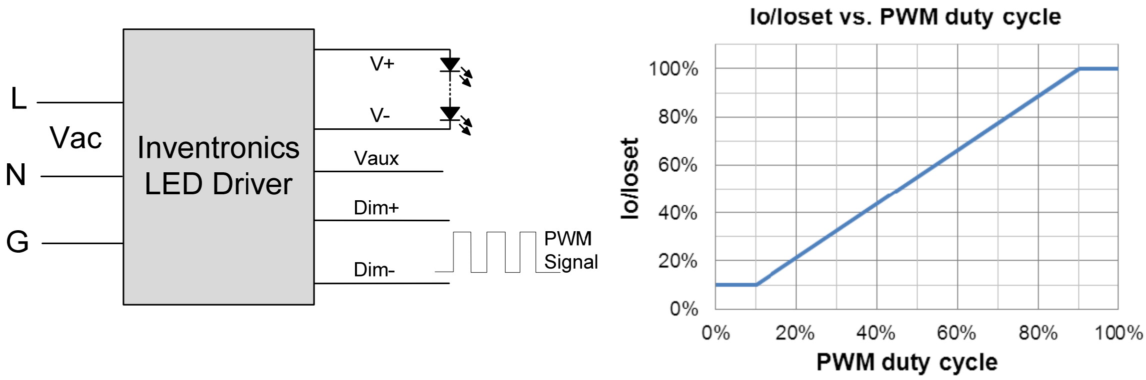
Implementation 4: Negative logic

Notes:

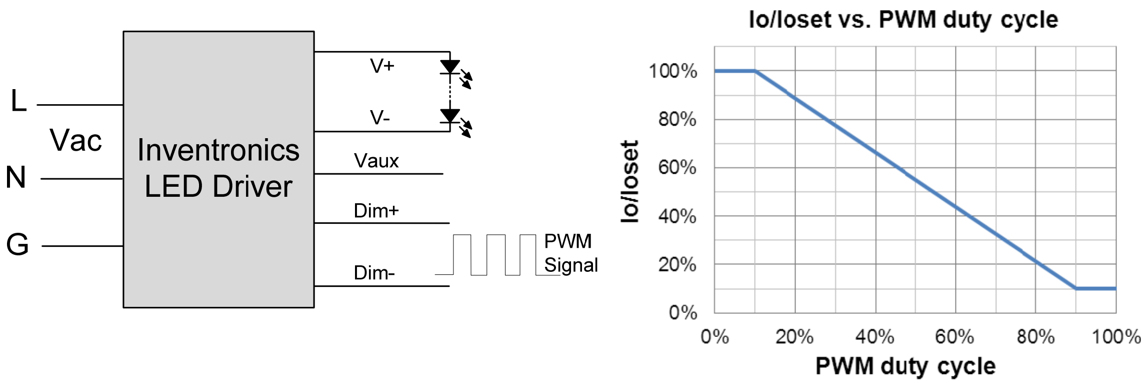
1. Do NOT connect Dim− to the output V− or V+, otherwise the driver will not work properly.
2. The dimmer can also be replaced by an active 1-10V voltage source signal or passive components like zener.
3. When 1-10V negative logic dimming mode and Dim+ is open, the driver will output minimum current.

● 10V PWM Dimming

The recommended implementation of the dimming control is provided below.



Implementation 5: Positive logic



Implementation 6: Negative logic

Notes:

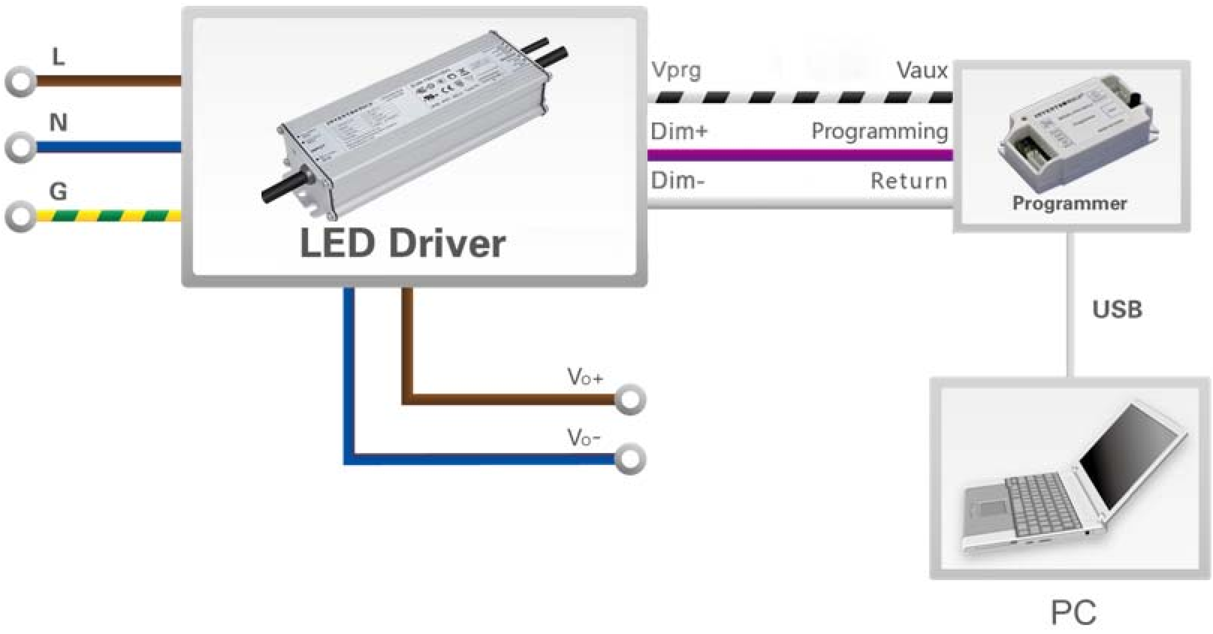
1. Do NOT connect Dim− to the output V− or V+, otherwise the driver will not work properly.
2. When PWM negative logic dimming mode and Dim+ is open, the driver will output minimum current.

- **Time Dimming**
  Time dimming control includes 3 kinds of modes, they are Self Adapting-Midnight, Self Adapting-Percentage and Traditional Timer.
  - **Self Adapting-Midnight**: Automatically adjusts the dimming curve based on the on-time of past two days (if difference <15 minutes), assuming that the center point of the dimming curve is midnight local time.
  - **Self Adapting-Percentage**: Automatically adjusts the on-time of each step by a constant percentage = (actual on-time for the past 2 days if difference <15 min) / (programmed on-time from the dimming curve).
  - **Traditional Timer**: Follows the programmed timing curve after power on with no changes.

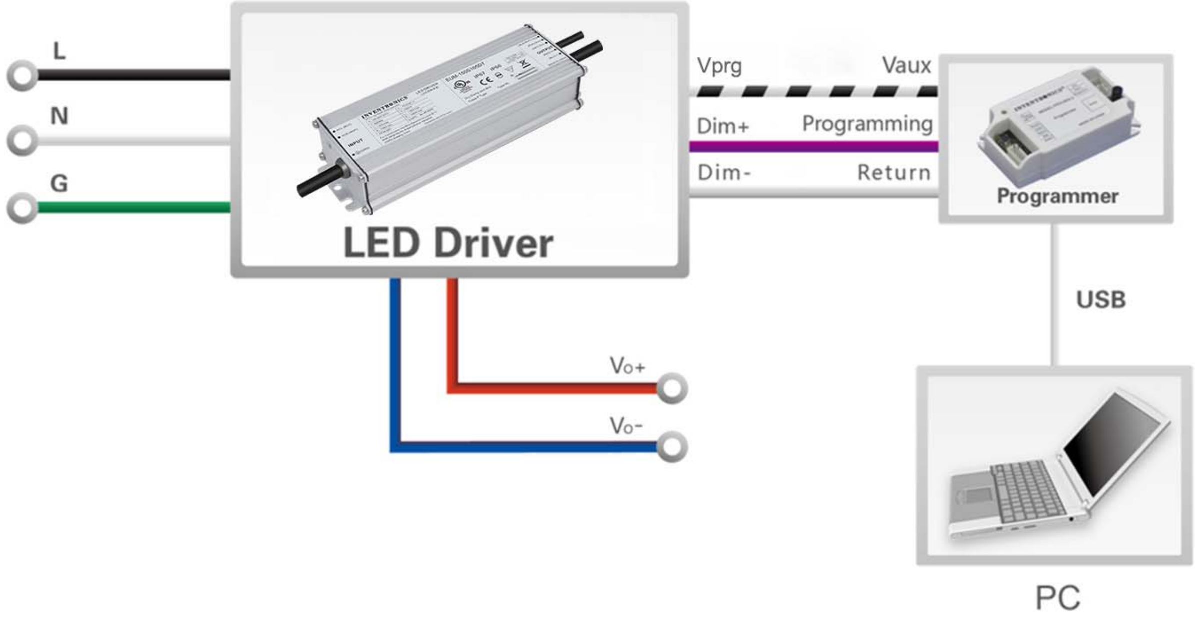
- **Output Lumen Compensation**
  Output Lumen Compensation (OLC) may be used to maintain constant light output over the life of the LEDs by driving them at a reduced current when new, then gradually increasing the drive current over time to counteract LED lumen degradation.

Programming Connection Diagram

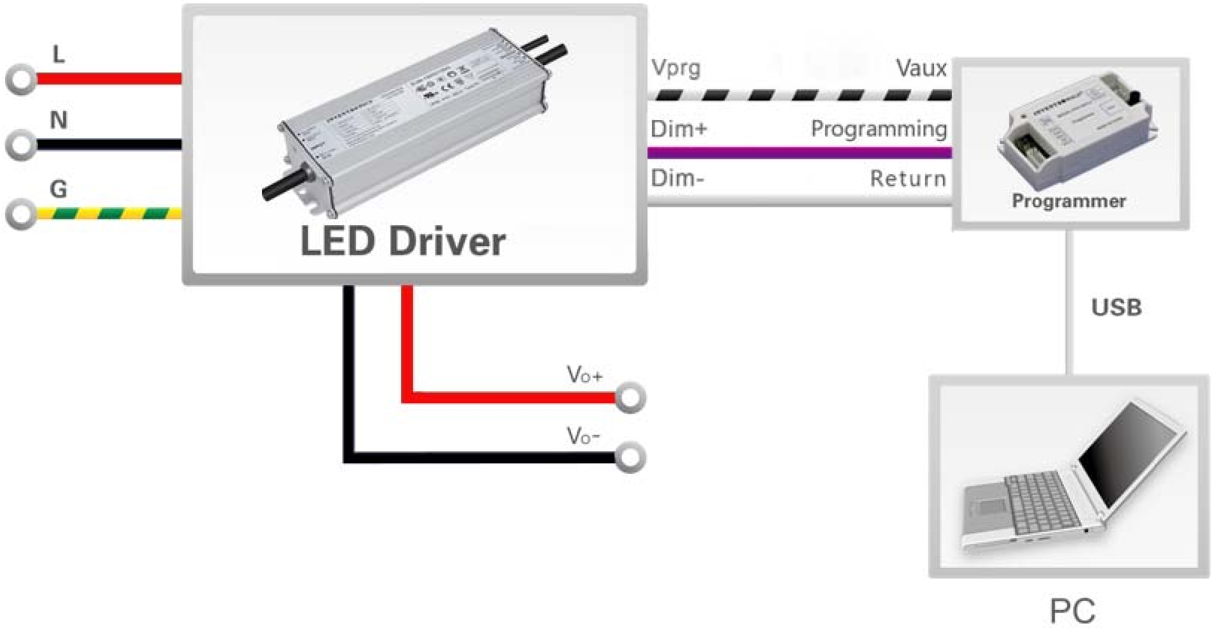
EUM-150SxxxDG



EUM-150SxxxDT



EUM-150SxxxDB

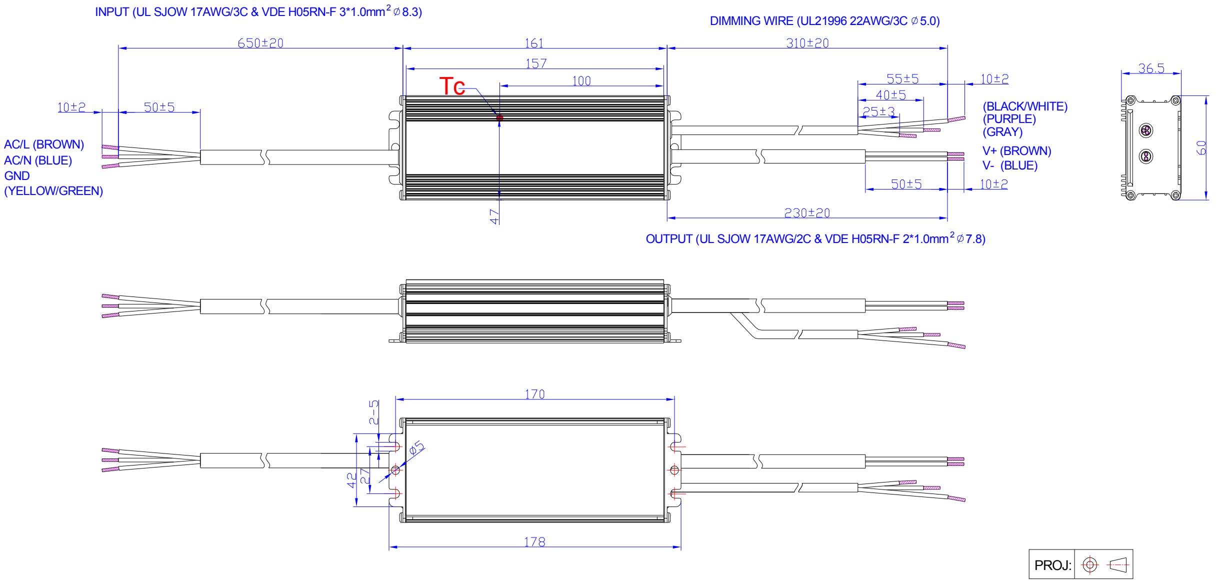


Note: The driver does not need to be powered on during the programming process.

- **Please refer to PRG-MUL2 (Programmer) datasheet for details.**

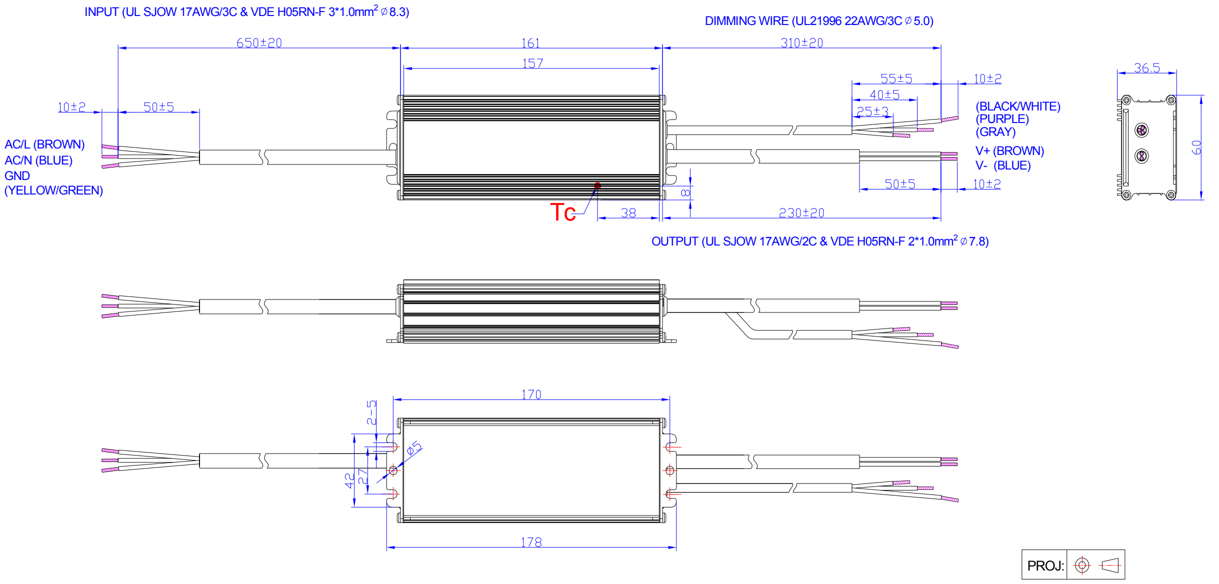
Mechanical Outline

EUM-150S105DG/EUM-150S150DG



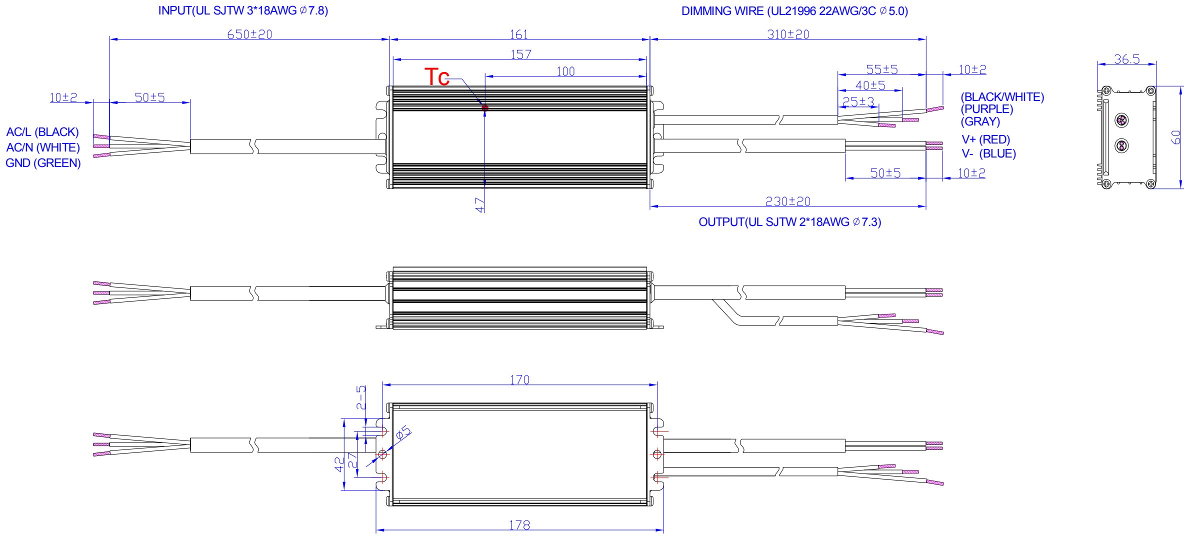
Unspecified tolerance:±1

EUM-150S210DG/EUM-150S420DG



Unspecified tolerance:±1

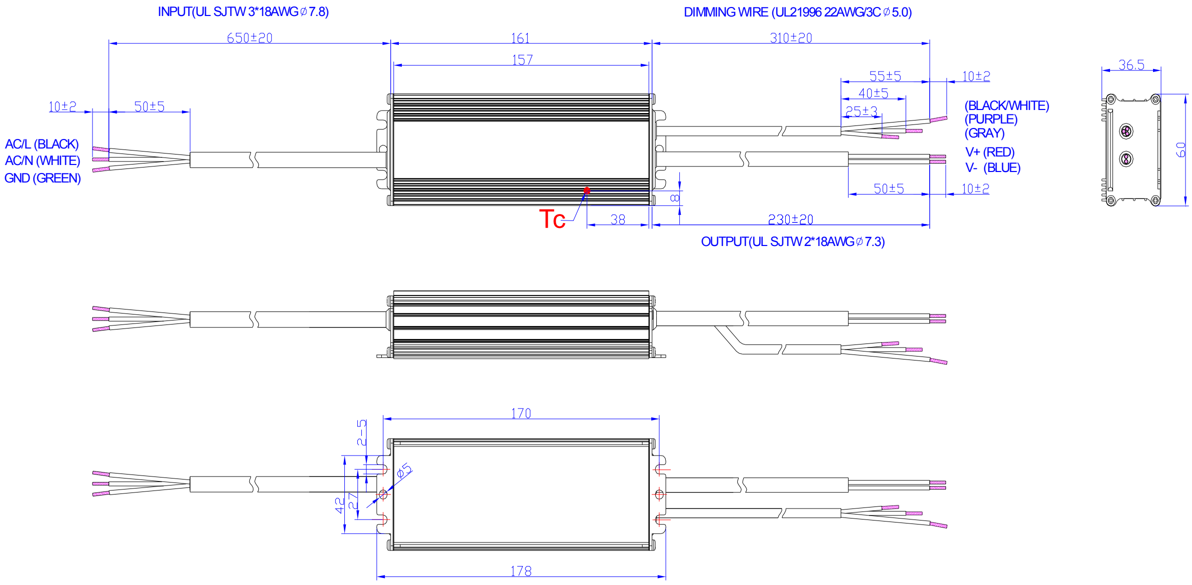
EUM-150S105DT/EUM-150S150DT



PROJ:

Unspecified tolerance:±1

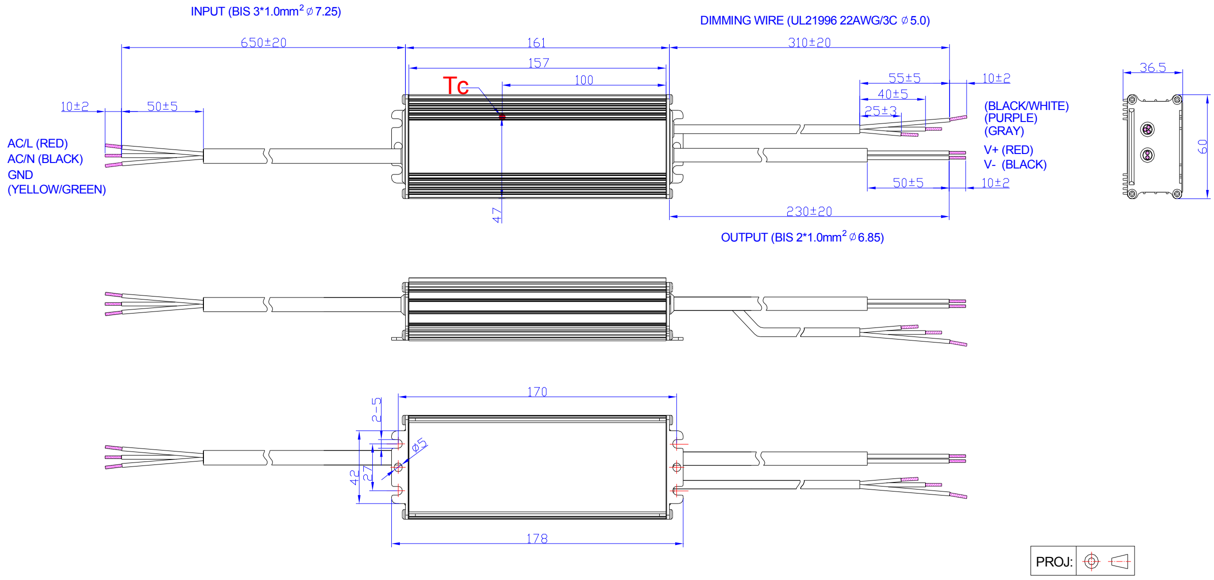
EUM-150S210DT/EUM-150S420DT



PROJ:

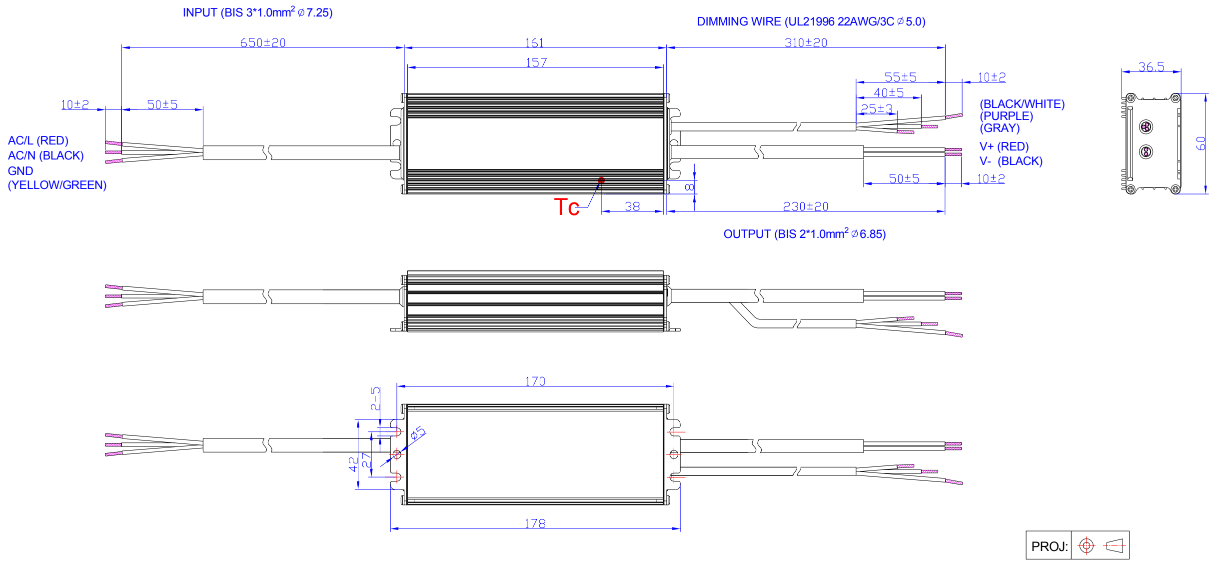
Unspecified tolerance:±1

EUM-150S105DB/EUM-150S150DB



Unspecified tolerance:±1

EUM-150S210DB/EUM-150S420DB



Unspecified tolerance:±1

RoHS Compliance

Our products comply with reference to RoHS Directive (EU) 2015/863 amending 2011/65/EU, calling for the elimination of lead and other hazardous substances from electronic products.

Revision History

Change Date	Rev.	Description of Change		
		Item	From	To
2021-03-09	A	Datasheets Release	/	/

Features

- Compact Metal Case with Excellent Thermal Performance
- Full Power at Wide Output Current Range (Constant Power)
- Adjustable Output Current (AOC) with Programmability
- Isolated 1-5V/1-10V/10V PWM/3-Timer-Modes Dimmable
- Output Lumen Compensation
- Input Surge Protection: DM 6kV, CM 10kV
- All-Around Protection: OVP, SCP, OTP
- IP66 / IP67 and UL Dry / Damp / Wet Location
- SELV Output
- TYPE HL, for use in a Class I, Division 2 hazardous (Classified) location
- 5 Years Warranty

Description

The *EUM-200SxxxDx* series is a 200W, constant-current, programmable IP67 LED driver that operates from 90-305Vac input with excellent power factor. It is created for many lighting applications including high bay, high mast and roadway, etc. The high efficiency of these drivers and compact metal case enables them to run cooler, significantly improving reliability and extending product life. To ensure trouble-free operation, protection is provided against input surge, output over voltage, short circuit, and over temperature.

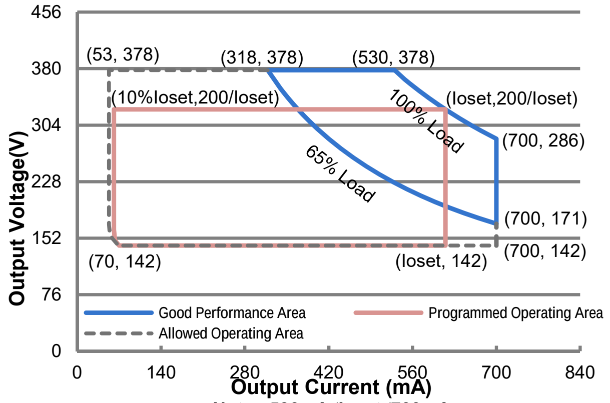
Models

Adjustable Output Current Range	Full-Power Current Range (1)	Default Output Current	Input Voltage Range(2)	Output Voltage Range	Max. Output Power	Typical Efficiency (3)	Typical Power Factor		Model Number (4) (5)
							120Vac	220Vac	
53-700mA	530-700mA	530 mA	90~305 Vac/ 127~300 Vdc	142~378 Vdc	200 W	94.0%	0.99	0.96	EUM-200S070Dx(6)
70-1050mA	700-1050mA	700 mA	90~305 Vac/ 127~300 Vdc	95~286 Vdc	200 W	93.5%	0.99	0.96	EUM-200S105Dx
105-1500mA	1050-1500mA	1050 mA	90~305 Vac/ 127~300 Vdc	67~190 Vdc	200 W	93.0%	0.99	0.96	EUM-200S150Dx
180-2800mA	1800-2800mA	2100 mA	90~305 Vac/ 127~300 Vdc	36~111 Vdc	200 W	92.5%	0.99	0.96	EUM-200S280Dx(7)
350-5600mA	3500-5600mA	4200 mA	90~305 Vac/ 127~300 Vdc	18 ~ 57 Vdc	200 W	92.0%	0.99	0.96	EUM-200S560Dx(7)

Notes: (1) Output current range with constant power at 200W.
(2) Certified input voltage range: UL, FCC 100-277Vac; otherwise 100-240Vac.
(3) Measured at 100% load and 220Vac input (see below "General Specifications" for details).
(4) x = G are UL Recognized, ENEC and CCC, etc. models; x = T are UL Class P models; x = B are BIS models.
(5) All the models are certificated to KS, except EUM-200S070Dx and EUM-200S105Dx.
(6) Only available with x = G, and only with ENEC, CE, CB and CCC certificates.
(7) SELV output.

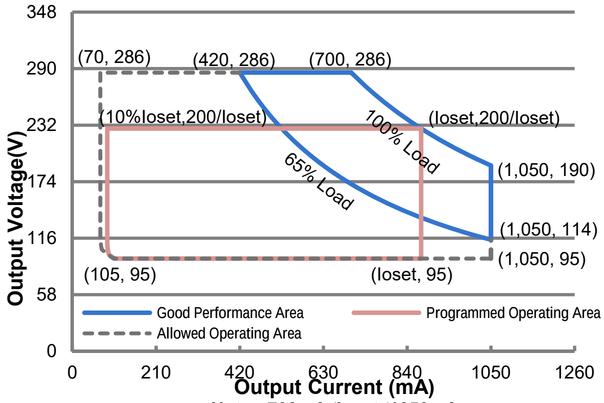
I-V Operation Area

EUM-200S070Dx



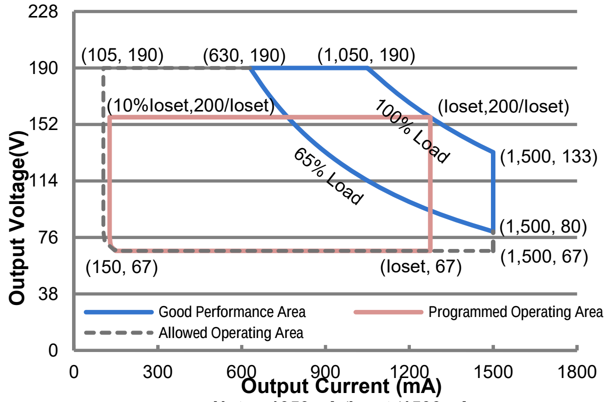
Note: 530mA≤Ioset≤700mA

EUM-200S105Dx



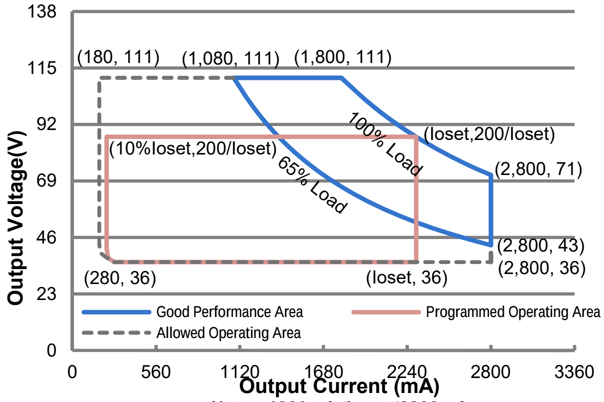
Note: 700mA≤Ioset≤1050mA

EUM-200S150Dx



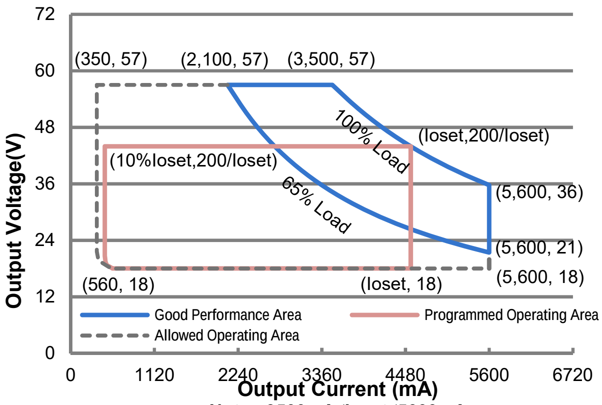
Note: 1050mA≤Ioset≤1500mA

EUM-200S280Dx



Note: 1800mA≤Ioset≤2800mA

EUM-200S560Dx



Note: 3500mA≤Ioset≤5600mA

Input Specifications

Parameter	Min.	Typ.	Max.	Notes
Input AC Voltage	90 Vac	-	305 Vac	
Input DC Voltage	127 Vdc	-	300 Vdc	
Input Frequency	47 Hz	-	63 Hz	
Leakage Current	-	-	0.75 MIU	UL8750; 277Vac/ 60Hz
	-	-	0.70 mA	IEC60598-1; 240Vac/ 60Hz
Input AC Current	-	-	2.00 A	Measured at 100% load and 120 Vac input.
	-	-	1.05 A	Measured at 100% load and 220 Vac input.
Inrush Current(I^2t)	-	-	4.20 A^2s	At 220Vac input, 25°C cold start, duration=848 μs, 10%Ipk-10%Ipk. See Inrush Current Waveform for the details.
PF	0.9	-	-	At 100-277Vac, 50-60Hz, 65%-100% Load (130-200W)
THD	-	-	20%	
THD	-	-	10%	At 220-240Vac, 50-60Hz, 75%-100% Load (150-200W)

Output Specifications

Parameter	Min.	Typ.	Max.	Notes
Output Current Tolerance	-5%Ioset	-	5%Ioset	At 100% load condition
Output Current Setting(Ioset) Range EUM-200S070Dx EUM-200S105Dx EUM-200S150Dx EUM-200S280Dx EUM-200S560Dx	 53 mA 70 mA 105 mA 180 mA 350 mA	 - - - - -	 700 mA 1050 mA 1500 mA 2800 mA 5600 mA	
Output Current Setting Range with Constant Power EUM-200S070Dx EUM-200S105Dx EUM-200S150Dx EUM-200S280Dx EUM-200S560Dx	 530 mA 700 mA 1050 mA 1800 mA 3500 mA	 - - - - -	 700 mA 1050 mA 1500 mA 2800 mA 5600 mA	
Total Output Current Ripple (pk-pk)	-	5%Iomax	10%Iomax	At 100% load condition. 20 MHz BW
Output Current Ripple at < 200 Hz (pk-pk)	-	2%Iomax	-	At 100% load condition. Only this component of ripple is associated with visible flicker.
Startup Overshoot Current	-	-	10%Iomax	At 100% load condition
No Load Output Voltage EUM-200S070Dx EUM-200S105Dx EUM-200S150Dx EUM-200S280Dx EUM-200S560Dx	 - - - - -	 - - - - -	 420 V 320 V 210 V 120 V 65 V	

Output Specifications (Continued)

Parameter	Min.	Typ.	Max.	Notes
Line Regulation	-	-	±0.5%	Measured at 100% load
Load Regulation	-	-	±1.5%	
Turn-on Delay Time	-	-	0.5 s	Measured at 120-277Vac input, 65%-100% Load
Temperature Coefficient of Ioset	-	0.03%/°C	-	Case temperature = 0°C ~Tc max

General Specifications

Parameter	Min.	Typ.	Max.	Notes
Efficiency at 120 Vac input:				Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
EUM-200S070Dx				
Io= 530 mA	89.0%	91.0%	-	
Io= 700 mA	89.5%	91.5%	-	
EUM-200S105Dx				
Io= 700 mA	88.5%	90.5%	-	
Io=1050 mA	89.0%	91.0%	-	
EUM-200S150Dx				
Io=1050 mA	88.5%	90.5%	-	
Io=1500 mA	88.5%	90.5%	-	
EUM-200S280Dx				
Io=1800 mA	87.0%	89.0%	-	
Io=2800 mA	87.0%	89.0%	-	
EUM-200S560Dx				
Io=3500 mA	87.5%	89.5%	-	
Io=5600 mA	87.0%	89.0%	-	
Efficiency at 220 Vac input:				Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
EUM-200S070Dx				
Io= 530 mA	92.0%	94.0%	-	
Io= 700 mA	92.0%	94.0%	-	
EUM-200S105Dx				
Io= 700 mA	91.5%	93.5%	-	
Io=1050 mA	91.5%	93.5%	-	
EUM-200S150Dx				
Io=1050 mA	91.0%	93.0%	-	
Io=1500 mA	91.0%	93.0%	-	
EUM-200S280Dx				
Io=1800 mA	90.5%	92.5%	-	
Io=2800 mA	90.0%	92.0%	-	
EUM-200S560Dx				
Io=3500 mA	90.0%	92.0%	-	
Io=5600 mA	89.5%	91.5%	-	

General Specifications (Continued)

Parameter	Min.	Typ.	Max.	Notes
Efficiency at 277 Vac input: EUM-200S070Dx Io= 530 mA Io= 700 mA EUM-200S105Dx Io= 700 mA Io=1050 mA EUM-200S150Dx Io=1050 mA Io=1500 mA EUM-200S280Dx Io=1800 mA Io=2800 mA EUM-200S560Dx Io=3500 mA Io=5600 mA	92.0% 92.5% 92.0% 92.0% 91.5% 91.5% 91.0% 90.5% 90.5% 90.0%	94.0% 94.5% 94.0% 94.0% 93.5% 93.5% 93.0% 92.5% 92.5% 92.0%	- - - - - - - - - -	Measured at 100% load and steady-state temperature in 25°C ambient; (Efficiency will be about 2.0% lower if measured immediately after startup.)
MTBF	-	267,000 Hours	-	Measured at 220Vac input, 80%Load and 25°C ambient temperature (MIL-HDBK-217F)
Lifetime	-	100,000 Hours	-	Measured at 220Vac input, 80%Load and 70°C case temperature; See lifetime vs. Tc curve for the details
Operating Case Temperature for Safety Tc_s	-40°C	-	+90°C	
Operating Case Temperature for Warranty Tc_w	-40°C	-	+80°C	Case temperature for 5 years warranty Humidity: 10%RH to 95%RH
Storage Temperature	-40°C	-	+85°C	Humidity: 5%RH to 95%RH
Dimensions Inches (L × W × H) Millimeters (L × W × H)		6.73 × 2.36 × 1.44 171 × 60 × 36.5		With mounting ear 7.40 × 2.36 × 1.44 188 × 60 × 36.5
Net Weight	-	780 g	-	

Dimming Specifications

Parameter		Min.	Typ.	Max.	Notes
Absolute Maximum Voltage on the Vdim (+) Pin		-20 V	-	20 V	
Source Current on Vdim (+)Pin		200 µA	300 µA	450 µA	Vdim(+) = 0 V
Dimming Output Range	EUM-200S070Dx EUM-200S105Dx EUM-200S150Dx EUM-200S280Dx EUM-200S560Dx	10%Ioset	-	Ioset	530 mA ⩽ Ioset ⩽ 700 mA 700 mA ⩽ Ioset ⩽ 1050 mA 1050 mA ⩽ Ioset ⩽ 1500 mA 1800 mA ⩽ Ioset ⩽ 2800 mA 3500 mA ⩽ Ioset ⩽ 5600 mA
	EUM-200S070Dx EUM-200S105Dx EUM-200S150Dx EUM-200S280Dx EUM-200S560Dx	53 mA 70 mA 105 mA 180 mA 350 mA	-	Ioset	53 mA ⩽ Ioset < 530 mA 70 mA ⩽ Ioset < 700 mA 105 mA ⩽ Ioset < 1050 mA 180 mA ⩽ Ioset < 1800 mA 350 mA ⩽ Ioset < 3500 mA
Recommended Dimming Range for 1-5V		0.25 V	-	4.75 V	Dimming mode set to 1-5V in PC interface.
Recommended Dimming Range for 1-10V		1 V	-	9 V	Default 1-10V dimming mode with positive logic.

Dimming Specifications (Continued)

Parameter	Min.	Typ.	Max.	Notes
PWM_in High Level	-	10V	-	
PWM_in Low Level	-	0V	-	
PWM_in Frequency Range	200 Hz	-	2 KHz	
PWM_in Duty Cycle	0%	-	100%	

Safety &EMC Compliance

Safety Category	Standard
UL/CUL	UL8750,CAN/CSA-C22.2 No. 250.13
ENEC & CE	EN 61347-1, EN 61347-2-13
UKCA	BS EN 61347-1, BS EN 61347-2-13
CB	IEC 61347-1, IEC 61347-2-13
CCC	GB 19510.1, GB 19510.14
PSE	J 61347-1, J 61347-2-13
KS	KS C 7655
BIS	IS 15885(Part2/Sec13)
EAC	ГОСТ Р МЭК 61347-1, ГОСТ IEC 61347-2-13
NOM	NOM-058-SCFI
EMI Standards	**Notes**
EN 55015/GB 17743/KN 15(1)	Conducted emission Test &Radiated emission Test
EN 61000-3-2/GB 17625.1	Harmonic current emissions
EN 61000-3-3	Voltage fluctuations & flicker
FCC Part 15(1)	ANSI C63.4 Class B
	This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: [1] this device may not cause harmful interference, and [2] this device must accept any interference received, including interference that may cause undesired Operation.
EMS Standards	**Notes**
EN 61000-4-2	Electrostatic Discharge (ESD): 8 kV air discharge, 4 kV contact discharge
EN 61000-4-3	Radio-Frequency Electromagnetic Field Susceptibility Test-RS
EN 61000-4-4	Electrical Fast Transient / Burst-EFT
EN 61000-4-5	Surge Immunity Test: AC Power Line: Differential Mode 6 kV, Common Mode 10 kV

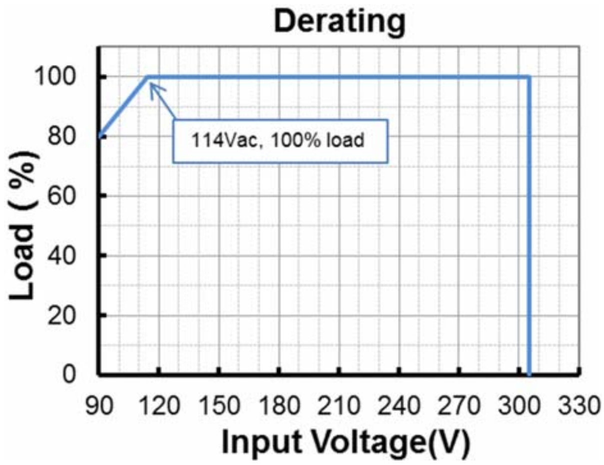
INVENTRONICS

Safety &EMC Compliance (Continued)

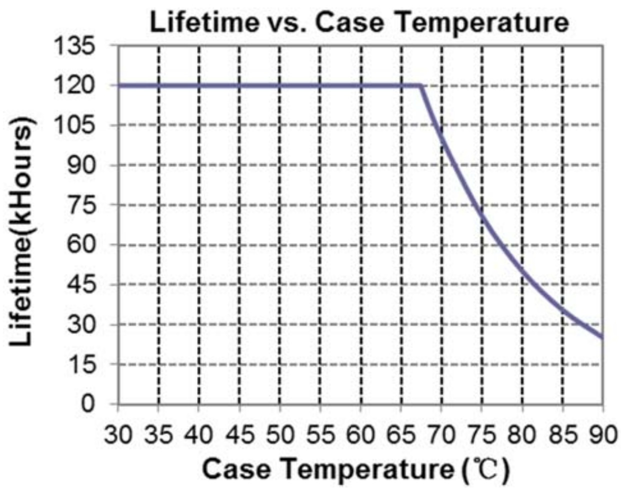
EMS Standards	Notes
EN 61000-4-6	Conducted Radio Frequency Disturbances Test-CS
EN 61000-4-8	Power Frequency Magnetic Field Test
EN 61000-4-11	Voltage Dips
EN 61547	Electromagnetic Immunity Requirements Applies To Lighting Equipment

Note: (1) This LED driver meets the EMI specifications above, but EMI performance of a luminaire that contains it depends also on the other devices connected to the driver and on the fixture itself.

Derating

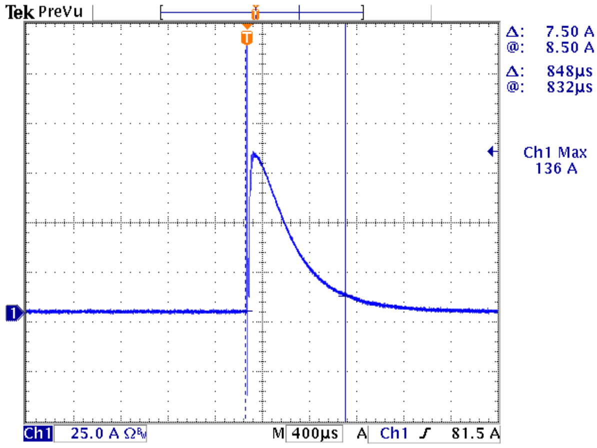


Lifetime vs. Case Temperature



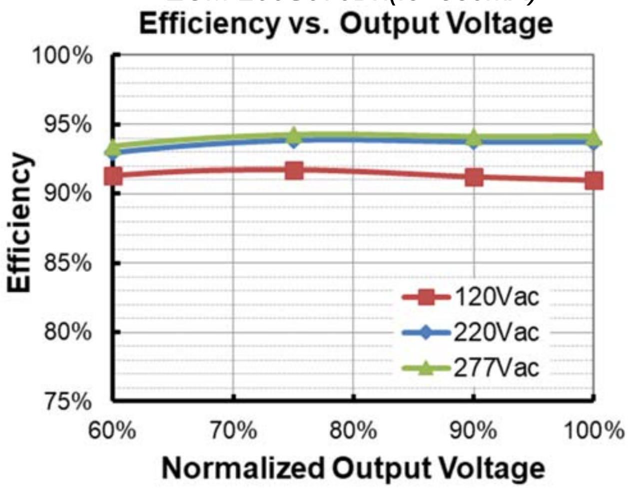
INVENTRONICS

Inrush Current Waveform

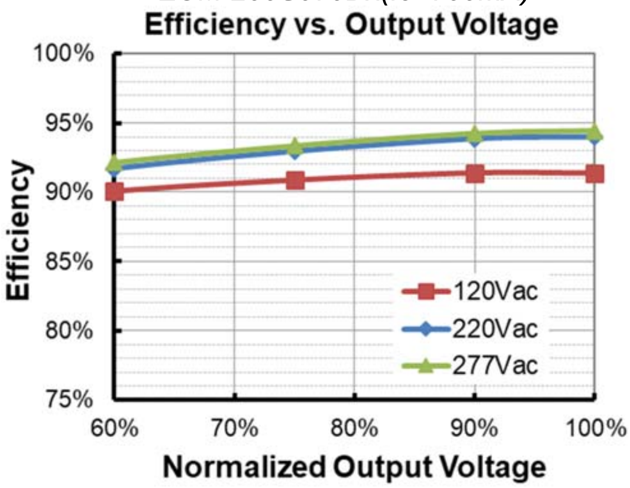


Efficiency vs. Load

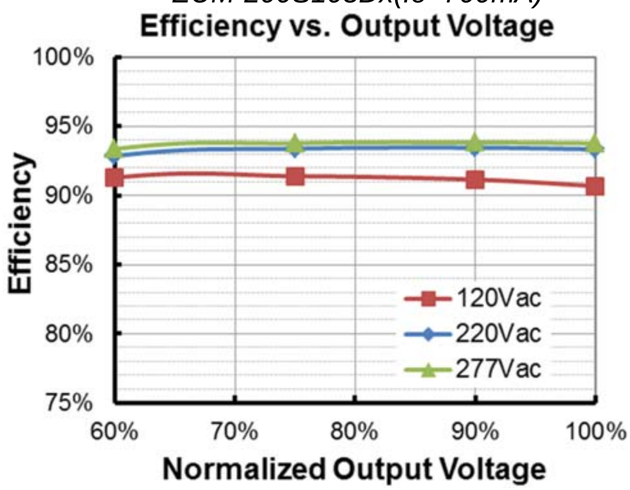
EUM-200S070Dx(Io=530mA)



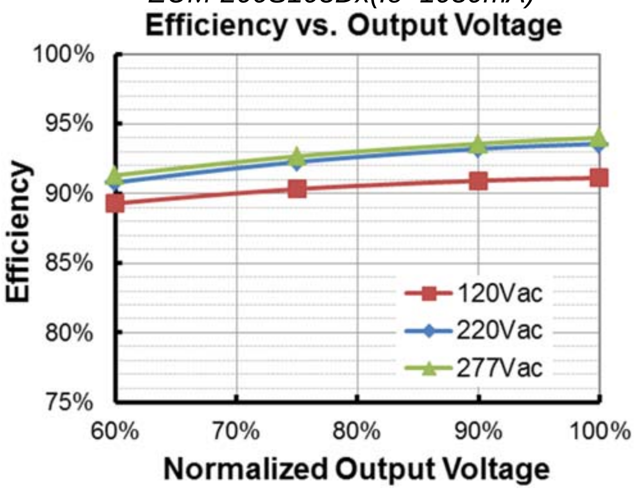
EUM-200S070Dx(Io=700mA)



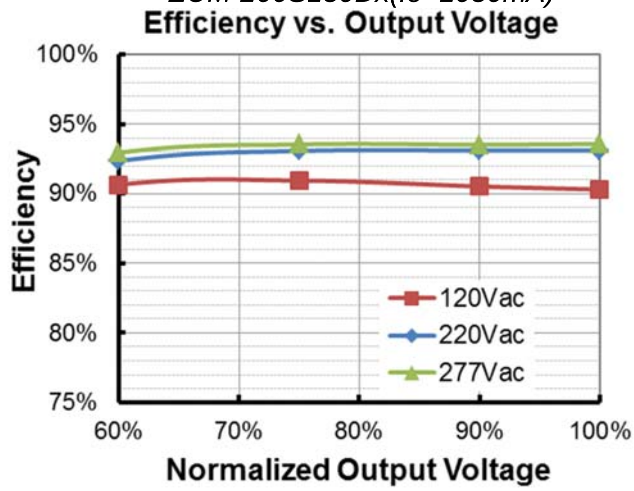
EUM-200S105Dx(Io=700mA)



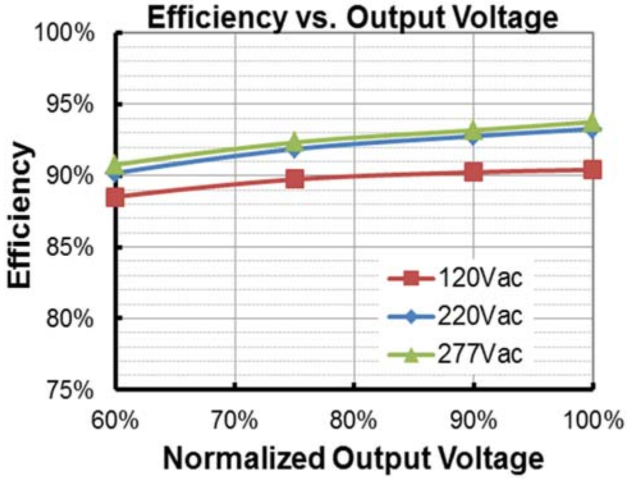
EUM-200S105Dx(Io=1050mA)



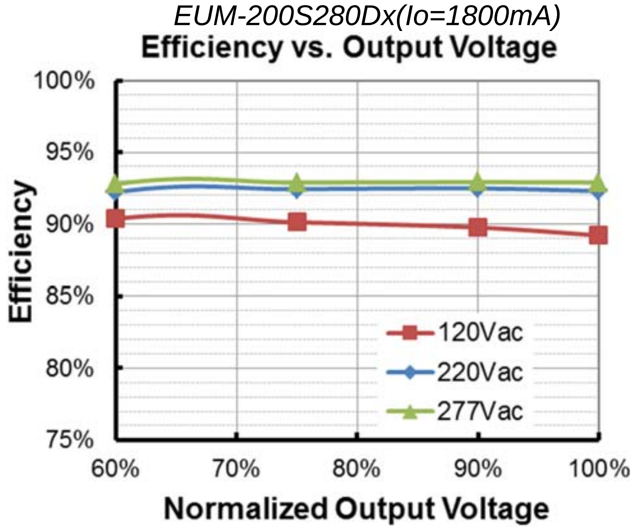
EUM-200S150Dx(Io=1050mA)



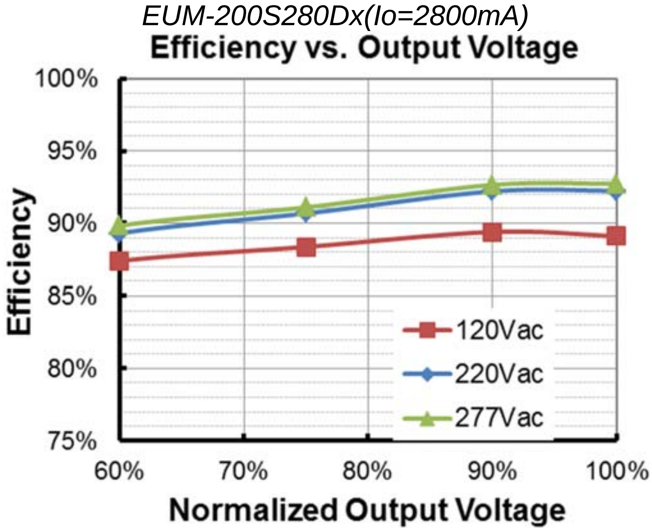
EUM-200S150Dx(Io=1500mA)



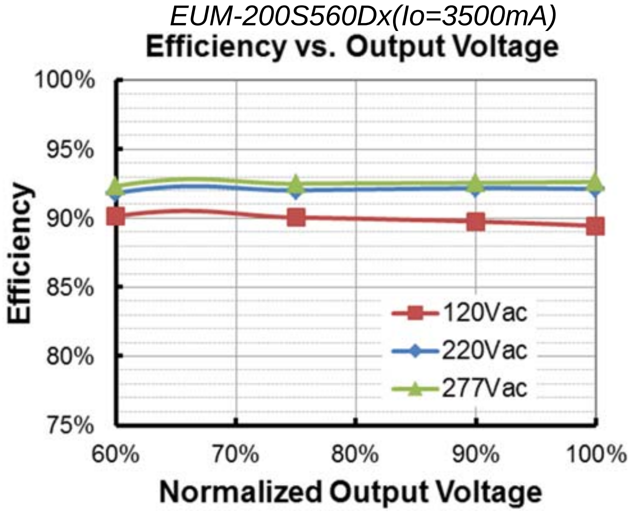
EUM-200S280Dx(Io=1800mA)



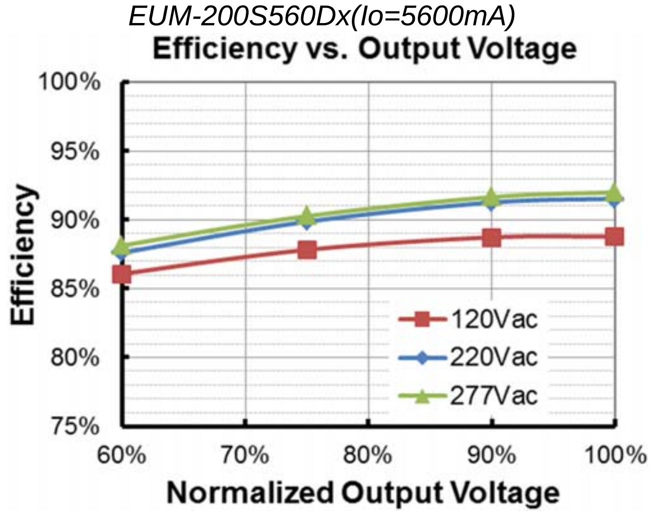
EUM-200S280Dx(Io=2800mA)



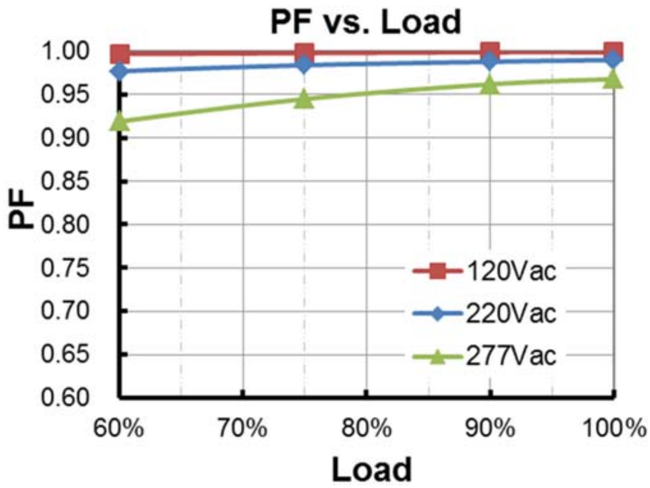
EUM-200S560Dx(Io=3500mA)



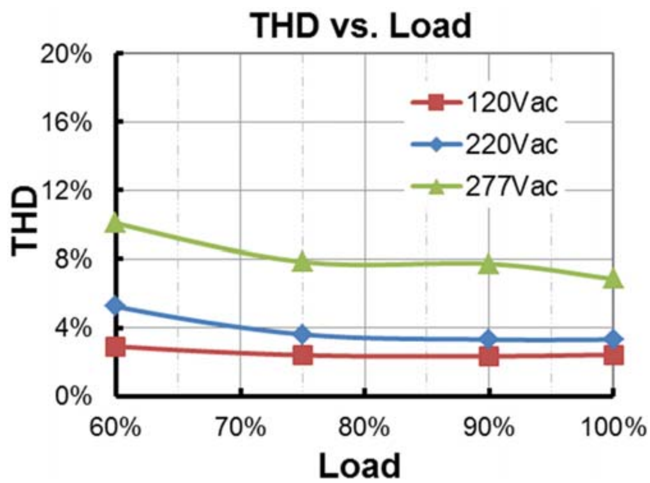
EUM-200S560Dx(Io=5600mA)



Power Factor



Total Harmonic Distortion



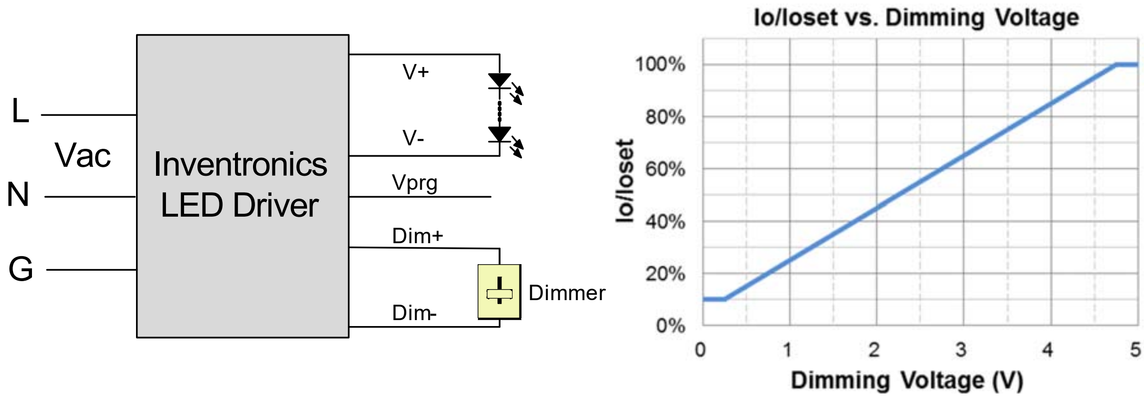
Protection Functions

Parameter	Notes
Over Temperature Protection	Decreases output current, returning to normal after over temperature is removed.
Short Circuit Protection	Auto Recovery. No damage will occur when any output is short circuited. The output shall return to normal when the fault condition is removed.
Over Voltage Protection	Limits output voltage at no load and in case the normal voltage limit fails.

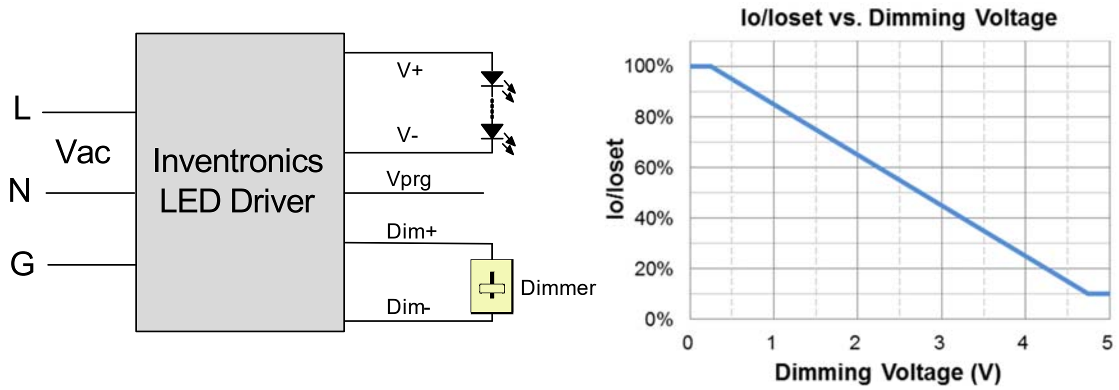
Dimming

- **1-5V Dimming**

The recommended implementation of the dimming control is provided below.



Implementation 1: Positive logic



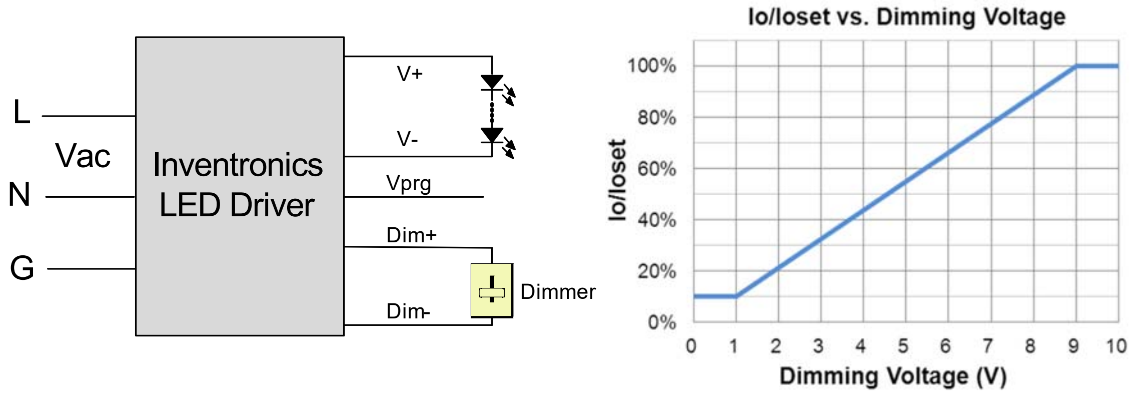
Implementation 2: Negative logic

Notes:

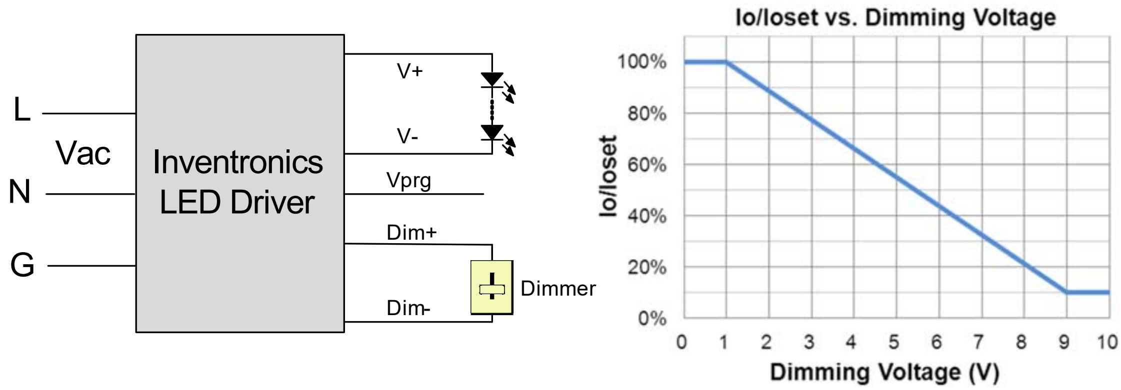
1. Do NOT connect Dim- to the output V- or V+, otherwise the driver will not work properly.
2. The dimmer can also be replaced by an active 1-5V voltage source signal or passive components like zener.
3. When 1-5V negative logic dimming mode and Dim+ is open, the driver will output maximum current.

● 1-10V Dimming

The recommended implementation of the dimming control is provided below.



Implementation 3: Positive logic



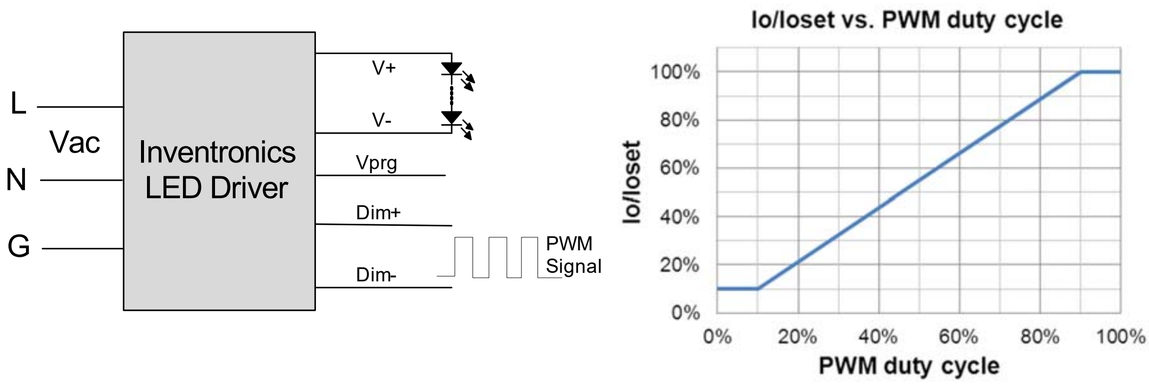
Implementation 4: Negative logic

Notes:

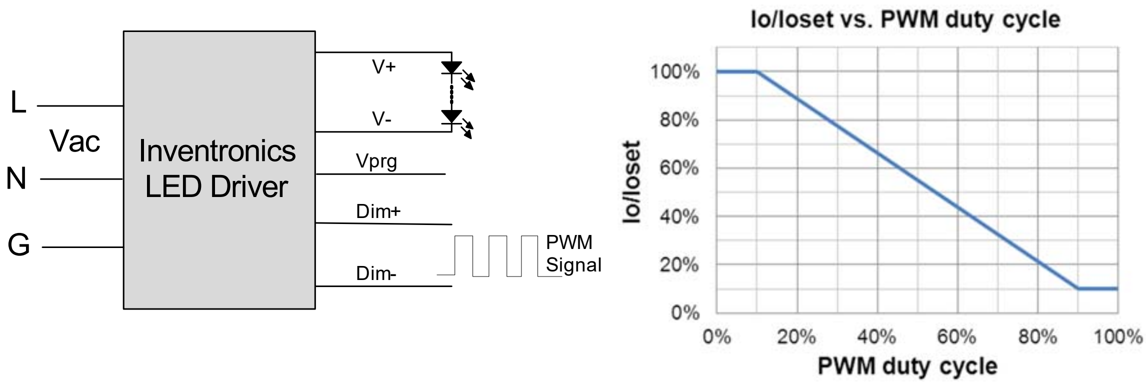
1. Do NOT connect Dim- to the output V- or V+, otherwise the driver will not work properly.
2. The dimmer can also be replaced by an active 1-10V voltage source signal or passive components like zener.
3. When 1-10V negative logic dimming mode and Dim+ is open, the driver will output minimum current.

● 10V PWM Dimming

The recommended implementation of the dimming control is provided below.



Implementation 5: Positive logic



Implementation 6: Negative logic

Notes:

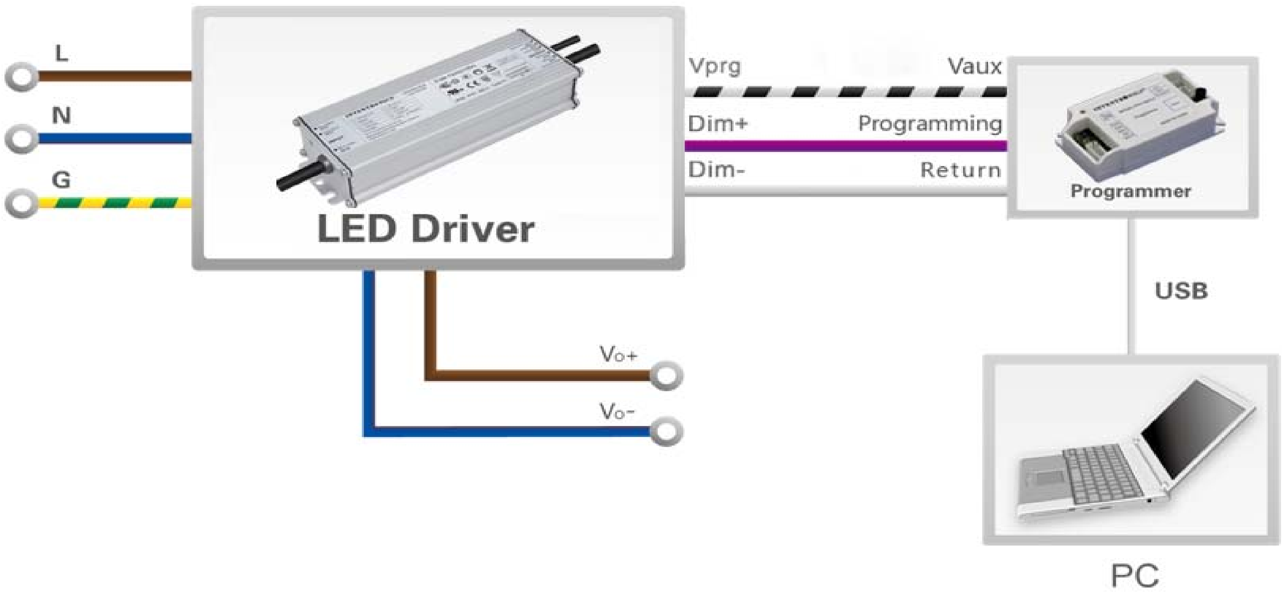
1. Do NOT connect Dim- to the output V- or V+, otherwise the driver will not work properly.
2. When PWM negative logic dimming mode and Dim+ is open, the driver will output minimum current.

- **Time Dimming**
  Time dimming control includes 3 kinds of modes, they are Self Adapting-Midnight, Self Adapting-Percentage and Traditional Timer.
  - **Self Adapting-Midnight**: Automatically adjusts the dimming curve based on the on-time of past two days (if difference <15 minutes), assuming that the center point of the dimming curve is midnight local time.
  - **Self Adapting-Percentage**: Automatically adjusts the on-time of each step by a constant percentage = (actual on-time for the past 2 days if difference <15 min) / (programmed on-time from the dimming curve).
  - **Traditional Timer**: Follows the programmed timing curve after power on with no changes.

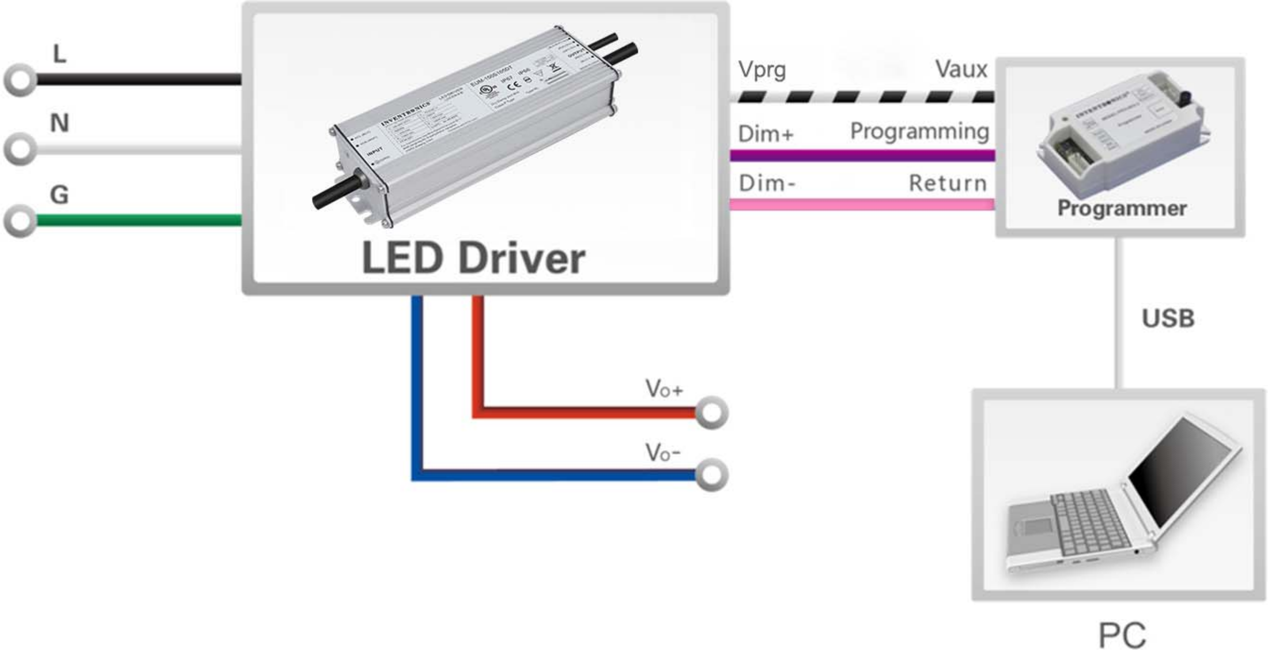
- **Output Lumen Compensation**
  Output Lumen Compensation (OLC) may be used to maintain constant light output over the life of the LEDs by driving them at a reduced current when new, then gradually increasing the drive current over time to counteract LED lumen degradation.

Programming Connection Diagram

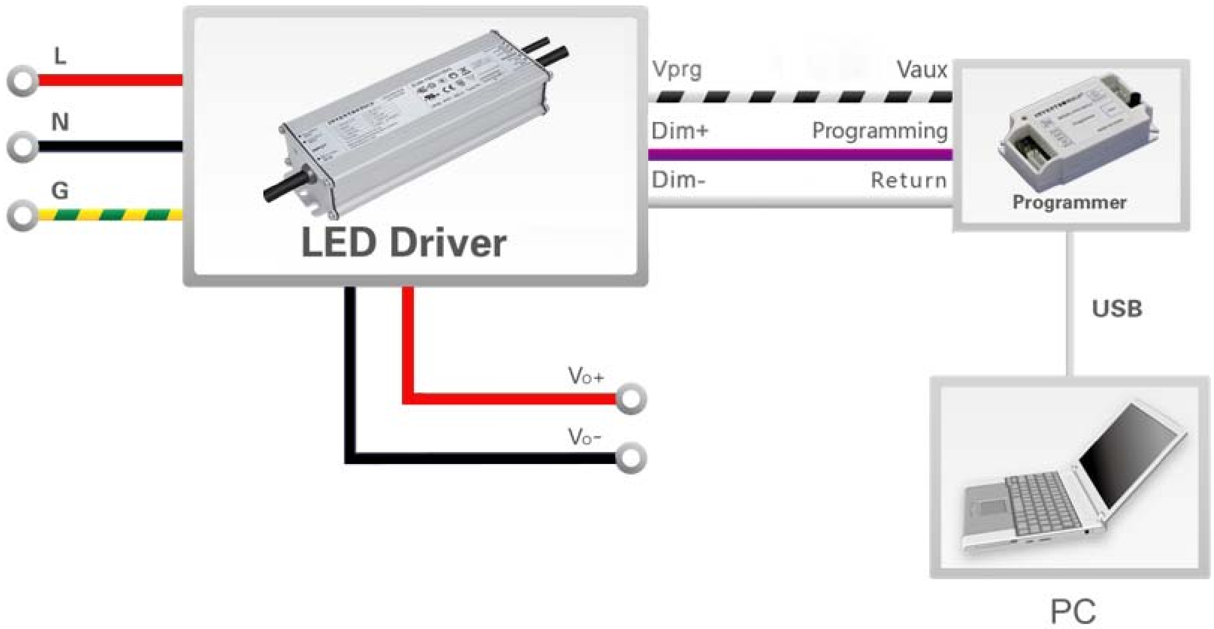
EUM-200SxxxDG



EUM-200SxxxDT



EUM-200SxxxDB

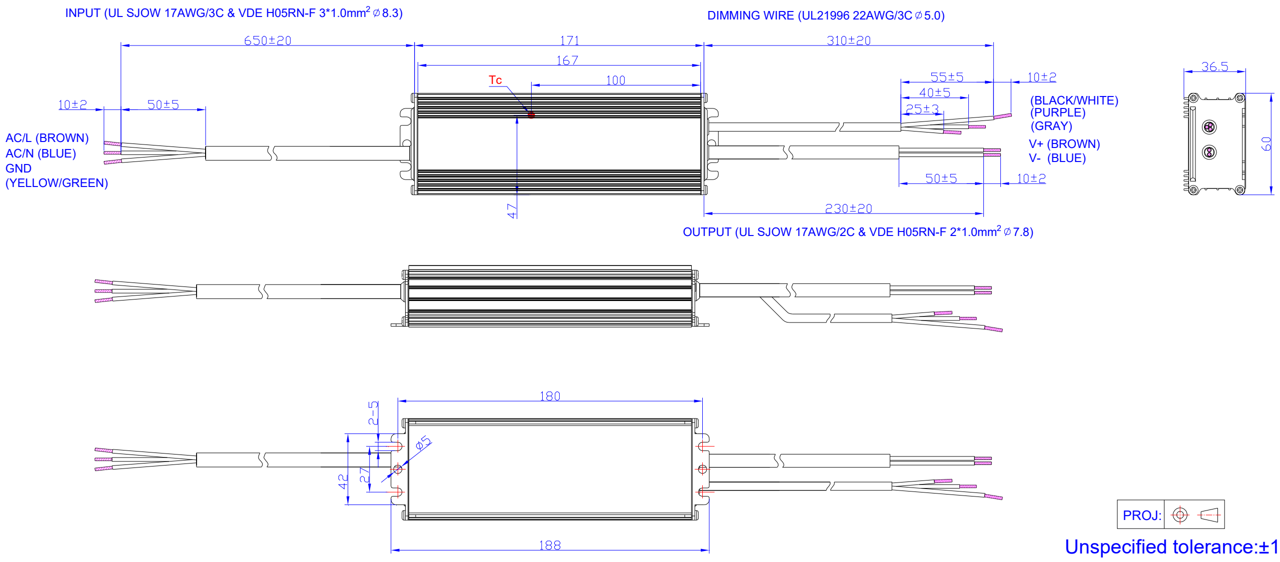


Note: The driver does not need to be powered on during the programming process.

- **Please refer to PRG-MUL2 (Programmer) datasheet for details.**

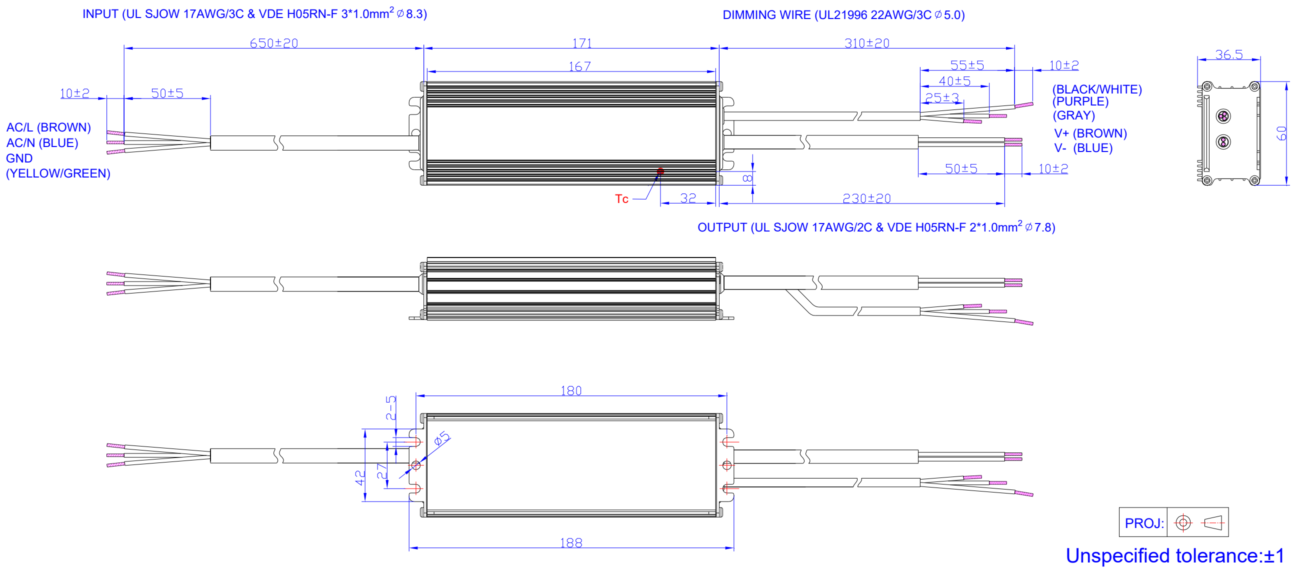
Mechanical Outline

EUM-200S070/105/150DG



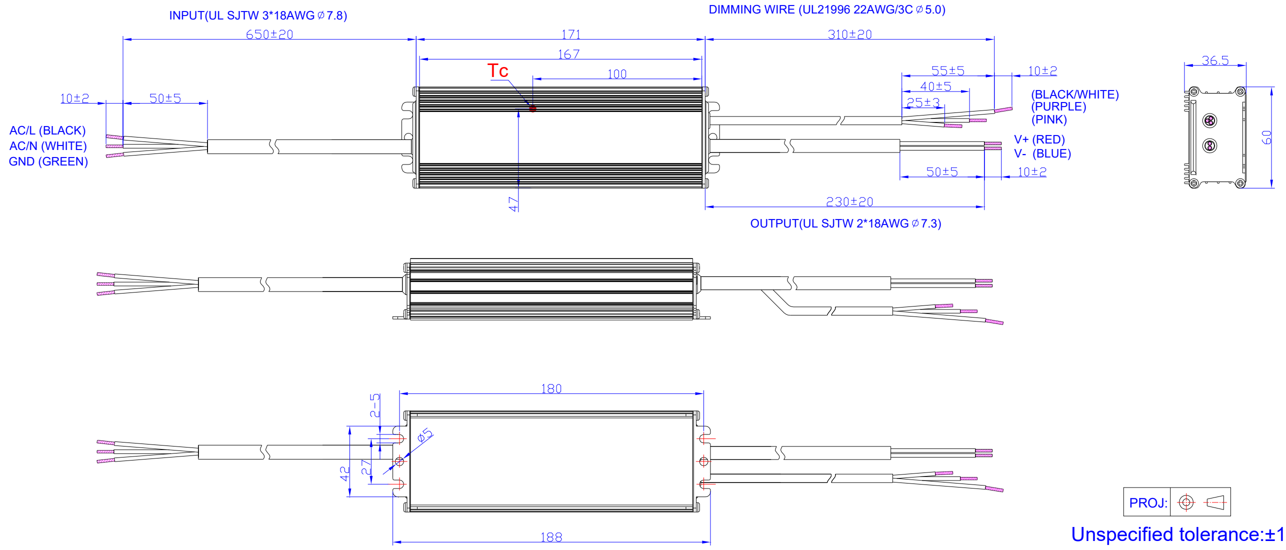
Unspecified tolerance:±1

EUM-200S280/560DG



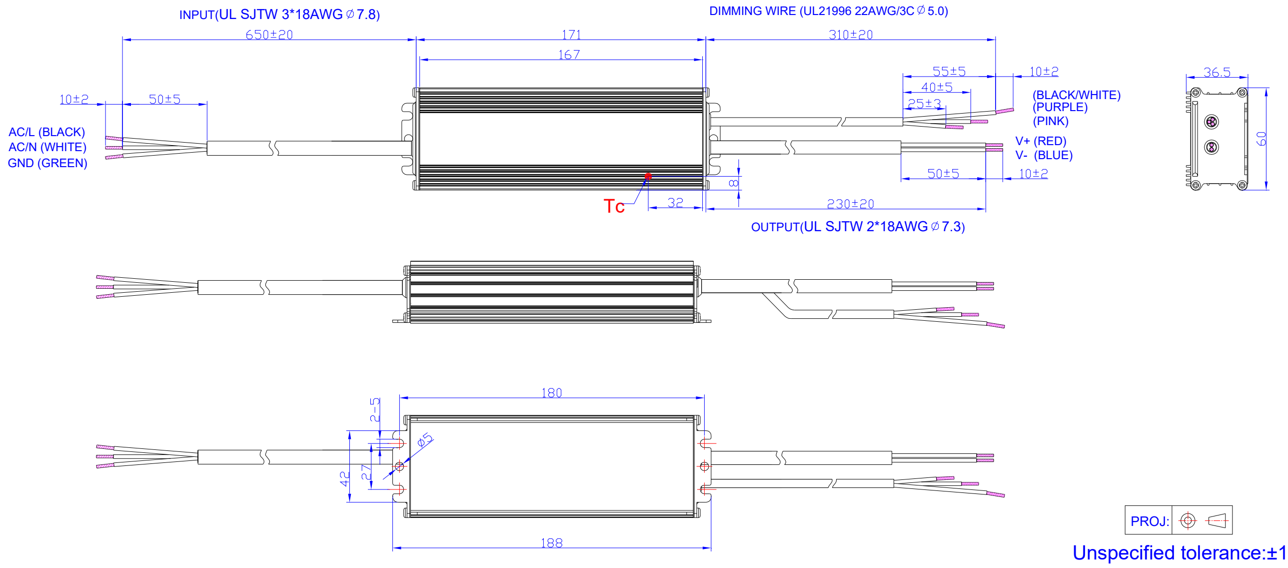
Unspecified tolerance:±1

EUM-200S105/150DT



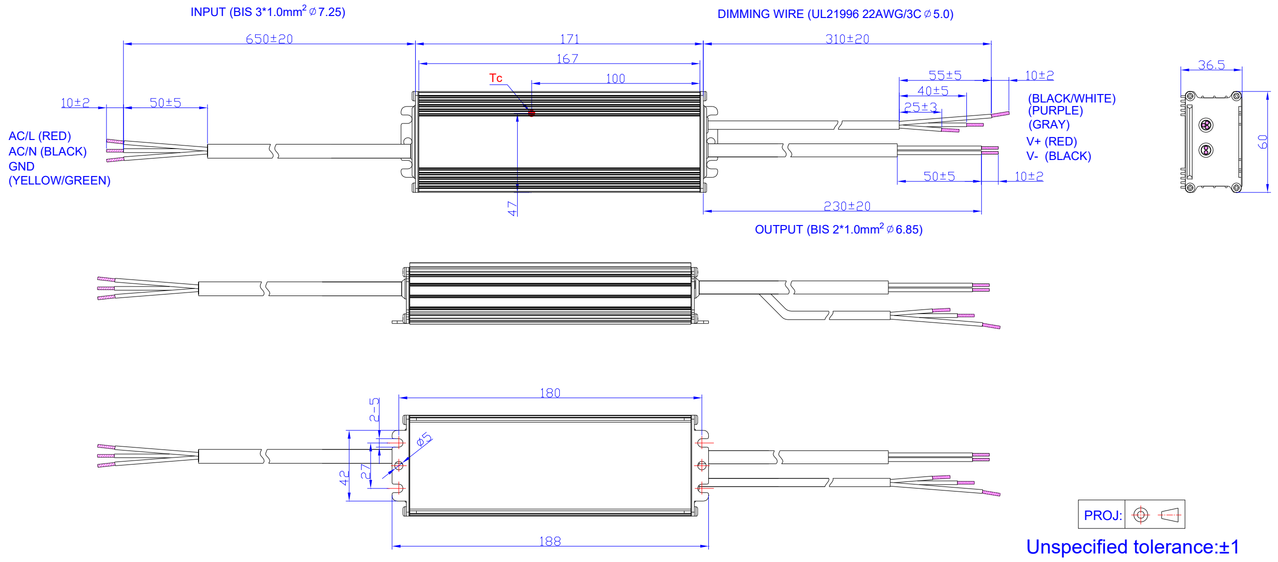
Unspecified tolerance:±1

EUM-200S280/560DT

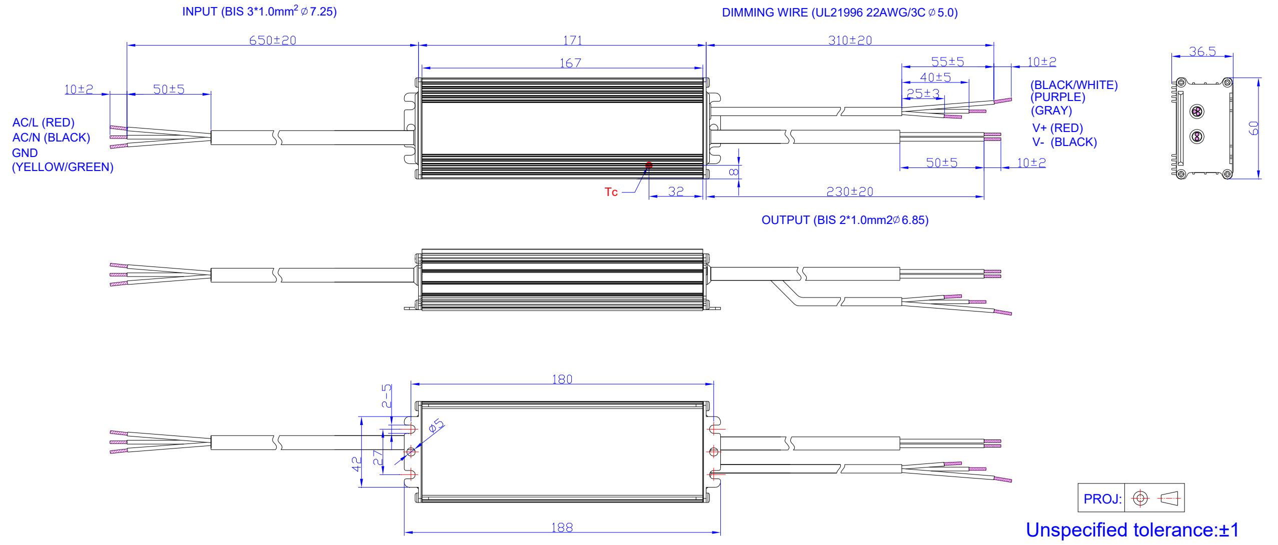


Unspecified tolerance:±1

EUM-200S105/150DB



EUM-200S280/560DB



RoHS Compliance

Our products comply with reference to RoHS Directive (EU) 2015/863 amending 2011/65/EU, calling for the elimination of lead and other hazardous substances from electronic products.

Revision History

Change Date	Rev.	Description of Change		
		Item	From	To
2021-03-09	A	Datasheets Release	/	/
2021-09-23	B	Models	EUM-200S070Dx	Added
		Models	Notes(6)、(7)	Added
		I-V Operation Area	EUM-200S070Dx	Added
		Output Current Setting(Ioset) Range	EUM-200S070Dx	Added
		Output Current Setting Range with Constant Power	EUM-200S070Dx	Added
		No Load Output Voltage	EUM-200S070Dx	Added
		Efficiency at 120 Vac input:	EUM-200S070Dx	Added
		Efficiency at 220 Vac input:	EUM-200S070Dx	Added
		Efficiency at 277 Vac input:	EUM-200S070Dx	Added
		Dimming Output Range	EUM-200S070Dx	Added
		Efficiency vs. Load	EUM-200S070Dx	Added
		Mechanical Outline	EUM-200S070DG	Added
2021-12-24	C	UKCA logo	/	Added
		General Specifications	Net Weight	Updated
		Safety &EMC Compliance	UKCA	Added
		Dimming Output Range	Note	Updated
		Programming Connection Diagram	EUM-200SxxxDT	Updated
		Mechanical Outline	/	Updated